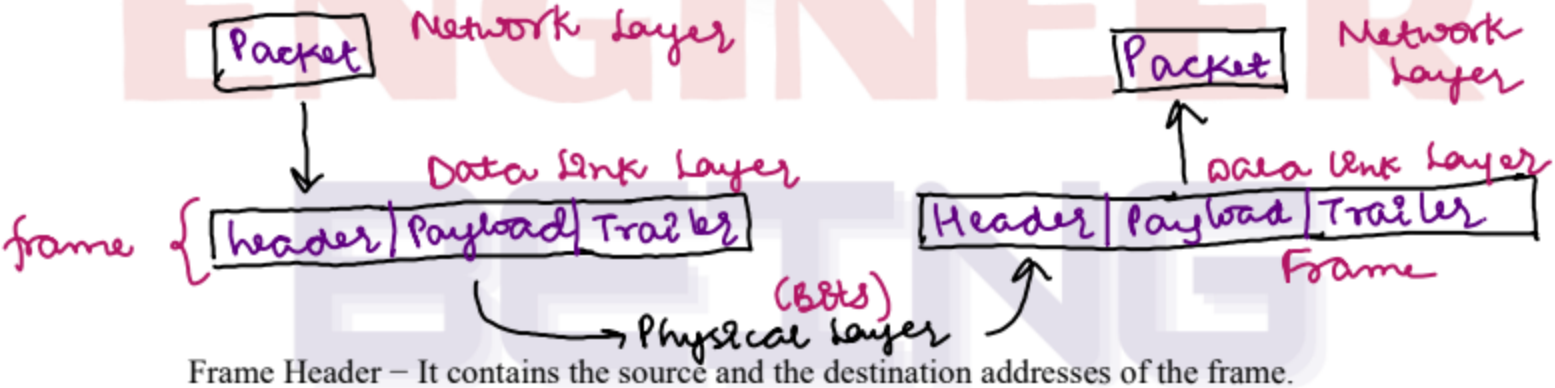
COMPUTER NETWORKS
Module 2 - DATA LINK LAYER

Link layer: Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols).
Medium Access Control and Local Area Networks: Channel allocation, Multiple access protocols, LAN standards, Link layer switches & bridges (learning bridge and spanning tree algorithms).

FUNCTIONS OF DLL

1. **Providing services to the network layer:**
   - Unacknowledged connectionless service.
     Appropriate for low error rate and real-time traffic.
     Ex: Ethernet
   - Acknowledged connectionless service.
     Useful in unreliable channels, WiFi. Ack/Timer/Resend
   - Acknowledged connection-oriented service.
     Guarantee frames are received exactly once and in the right order. Appropriate over long, unreliable links such as a satellite channel or a long-distance telephone circuit

2. Framing: Frames are the streams of bits received from the network layer into manageable data units. This division of the stream of bits is done by the Data Link Layer.



Frame Header − It contains the source and the destination addresses of the frame.
Payload field − It contains the message to be delivered.
Trailer − It contains the error detection and error correction b

3. Physical Addressing: The Data Link layer adds a header to the frame in order to define the physical address of the sender or receiver of the frame, if the frames are to be distributed to different systems on the network.

4. Flow Control: A receiving node can receive the frames at a faster rate than it can process the frame. Without flow control, the receiver's buffer can overflow, and frames can get lost. To overcome this problem, the data link layer uses the flow control to prevent the sending node on one side of the link from overwhelming the receiving node on another side of the link. This prevents traffic jams at the receiver side.

5. Error Control: Error control is achieved by adding a trailer at the end of the frame. Duplication of frames is also prevented by using this mechanism. Data Link Layers adds a mechanism to prevent duplication of frames.

- Error detection: Errors can be introduced by signal attenuation and noise. Data Link Layer protocol provides a mechanism to detect one or more errors. This is achieved by adding error detection bits in the frame and then receiving nodes can perform an error check.

- Error correction: Error correction is similar to the Error detection, except that the receiving node not only detects the errors but also determines where the errors have occurred in the frame.

6. Access Control: Protocols of this layer determine which of the devices has control over the link at any given time, when two or more devices are connected to the same link.

TYPES OF FRAMING

Framing can be of two types:

Fixed-sized Framing**:** Here the size of the frame is fixed and so the frame length acts as delimiter of the frame. Consequently, it does not require additional boundary bits to identify the start and end of the frame. Example − ATM cells.

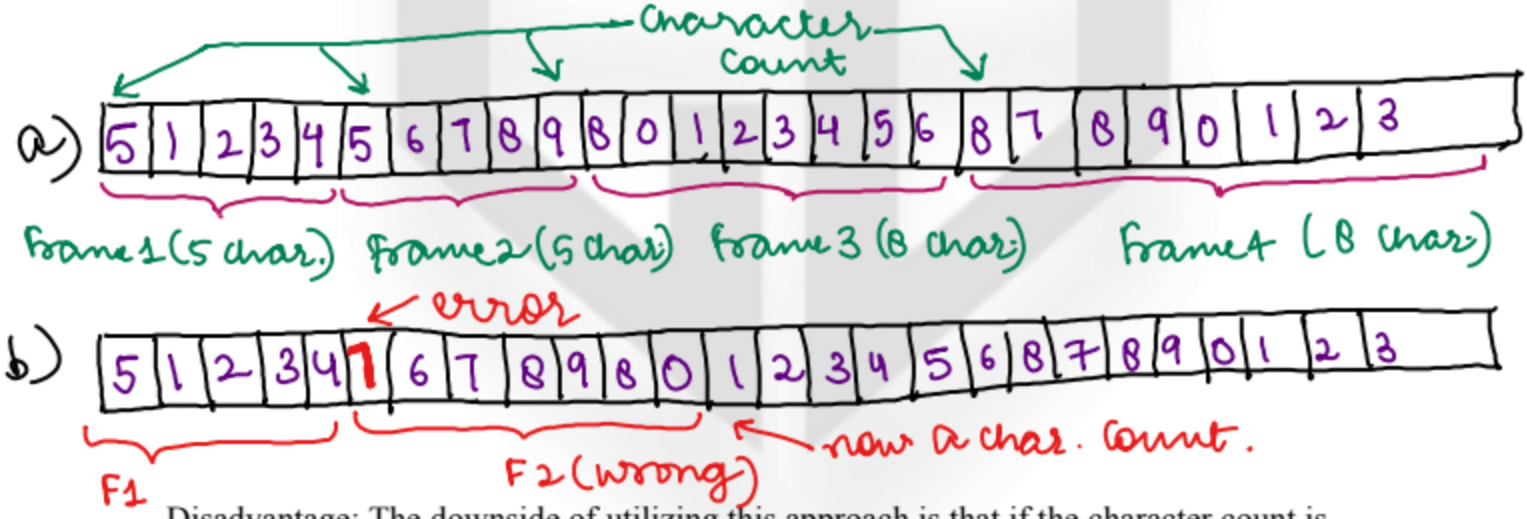
Variable – Sized Framing: Here, the size of each frame to be transmitted may be different. So additional mechanisms are kept marking the end of one frame and the beginning of the next frame. It is used in local area networks.

METHODS OF FRAMING

1. Character count.
2. Flag bytes with byte stuffing.
3. Starting and ending flags, with bit stuffing.
4.Physical Layer Coding Violation

CHARACTER COUNT / BYTE COUNT

The Character count method uses a field in the header to specify the number of characters in the frame. When the data link layer at the destination sees the character count, it knows how many characters follow and hence where the end of the frame is.



Disadvantage: The downside of utilizing this approach is that if the character count is interrupted or corrupted in any way during transmission, the destination or receiver may lose synchronization. The destination or receiver may also be unable to discover or identify the start of the following frame.

FLAG BYTES WITH BYTE STUFFING

- This method gets around the problem of resynchronization after an error by having each frame start and end with special bytes.
- This protocol has used the same byte in the starting and ending frame, called a flag byte.
- If the receiver ever loses synchronization, it can just search for the flag byte to find the end of the current frame.
- Two consecutive flag bytes indicate the end of one frame and start of the next one.



Original characters → After stuffing.

A FLAG B → A ESC FLAG B

A ESC B → A ESC ESC B

A ESC FLAG B → A ESC ESC ESC FLAG B

A ESC ESC B → A ESC ESC ESC ESC B

Try yourself - A B ESC C ESC FLAG FLAG D

FLAG BYTE WITH BIT STUFFING

Each frame begins and ends with a special bit pattern, 01111110 (in fact, a flag byte). Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream.

A 01111110 B → A 011111010 B → A 01111110 B

(stuffed 0)

Ex-

0110111111111111111110010

0110111110111110111110010

(stuffed 0 after 5 1's)

PHYSICAL LAYER CODING VIOLATION

This method of framing is only applicable to networks in which the encoding on the physical medium contains some redundancy. For example, some LANs encode 1 bit of data by using 2 physical bits. Normally, a 1 bit is a high-low pair and a 0 bit is a low-high pair. The scheme means that every data bit has a transition in the middle, making it easy for the receiver to locate the bit boundaries. The combinations high- high and low-low are not used for data but are used for delimiting frames in some protocols.

Ques: The following character encoding is used in a data link protocol: A: 01000111; B: 11100011; FLAG: 01111110; ESC: 11100000 Show the bit sequence transmitted (in binary) for the four-character frame: A B ESC FLAG when each of the following framing methods are used: (a) Character count (b) Flag bytes with byte stuffing. (c) Starting and ending flag bytes, with bit stuffing.

Ques: Given the output after byte-stuffing:
FLAG A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D FLAG.
What is the original data?

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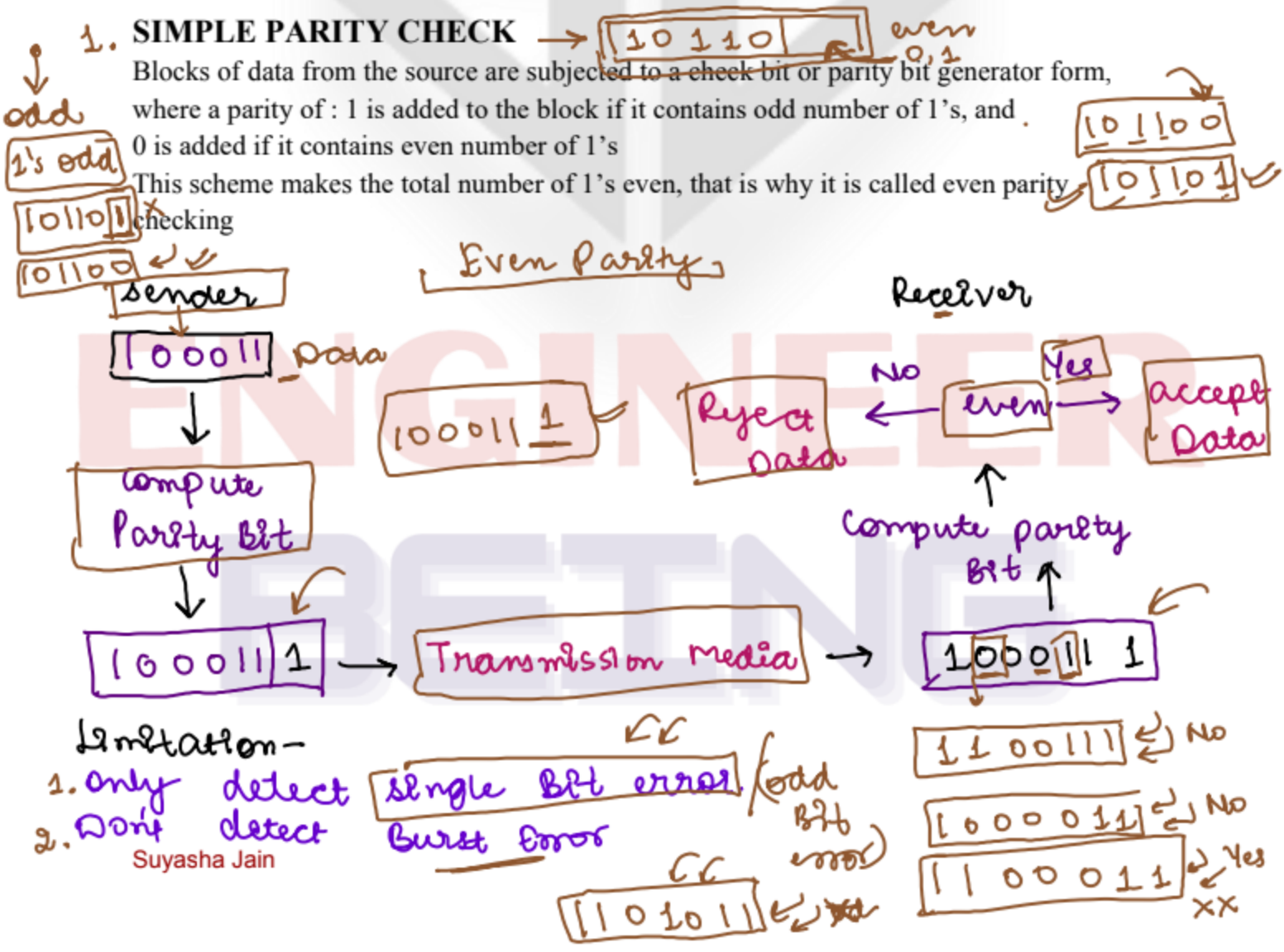
ERROR CONTROL

Error Detection	Error Correction
1. Receiver can know an error has occurred has occurred and (but not which error)	1. Receiver can know an error is also able to correct it.
2.useful for reliable links such as fiber optic cable noisy/unreliable/wireless channels.	2. useful for
3.Eg: Parity, Checksums, Cyclic Redundancy Check (CRC)	3.Eg: Hamming code,

1. SIMPLE PARITY CHECK

Blocks of data from the source are subjected to a check bit or parity bit generator form, where a parity of : 1 is added to the block if it contains odd number of 1's, and 0 is added if it contains even number of 1's

This scheme makes the total number of 1's even, that is why it is called even parity checking

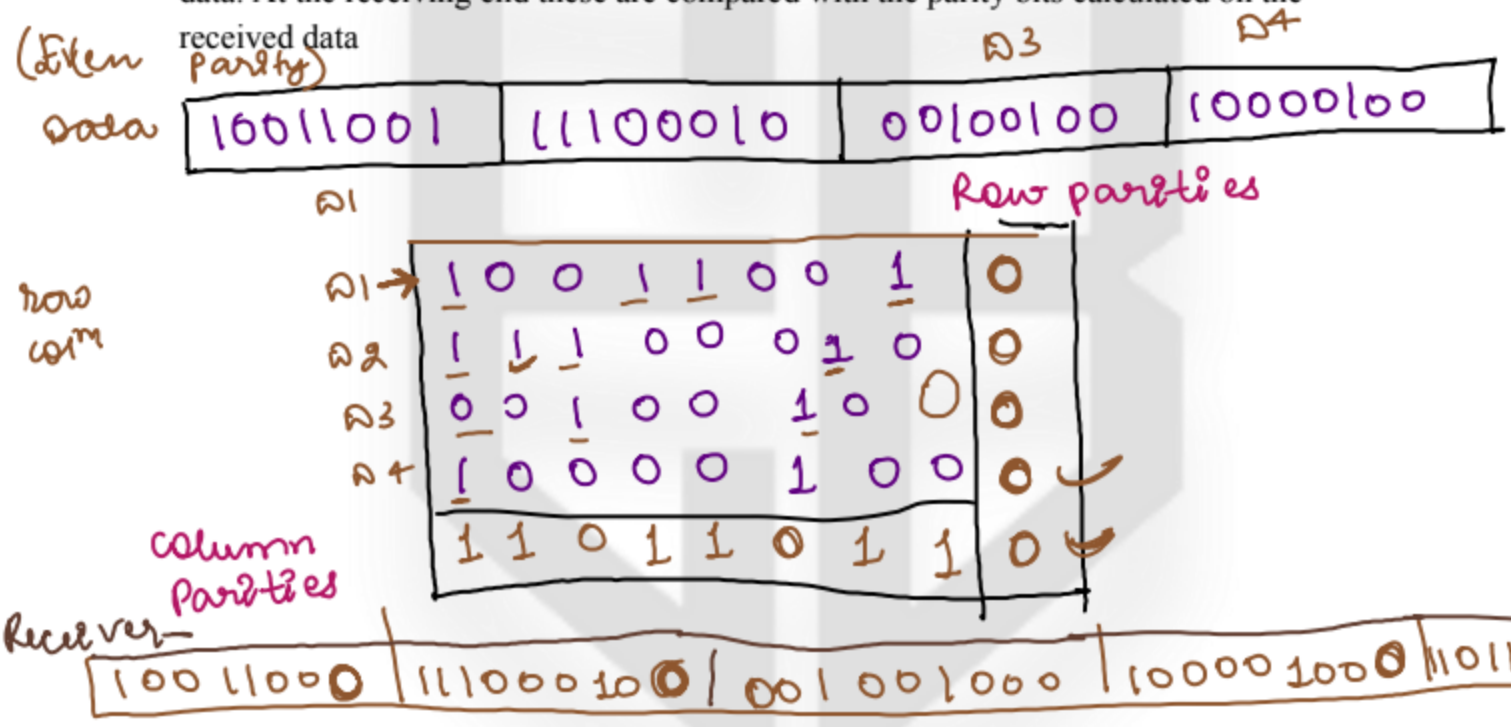


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disadv. if 2 bits are in exactly same position in other data are damaged then we'll not detect.

2. TWO-DIMENSIONAL PARITY CHECK

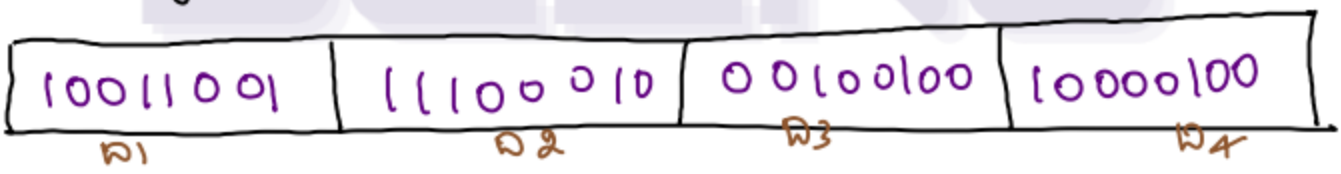
Parity check bits are calculated for each row, which is equivalent to a simple parity check bit. Parity check bits are also calculated for all columns, then both are sent along with the data. At the receiving end these are compared with the parity bits calculated on the received data



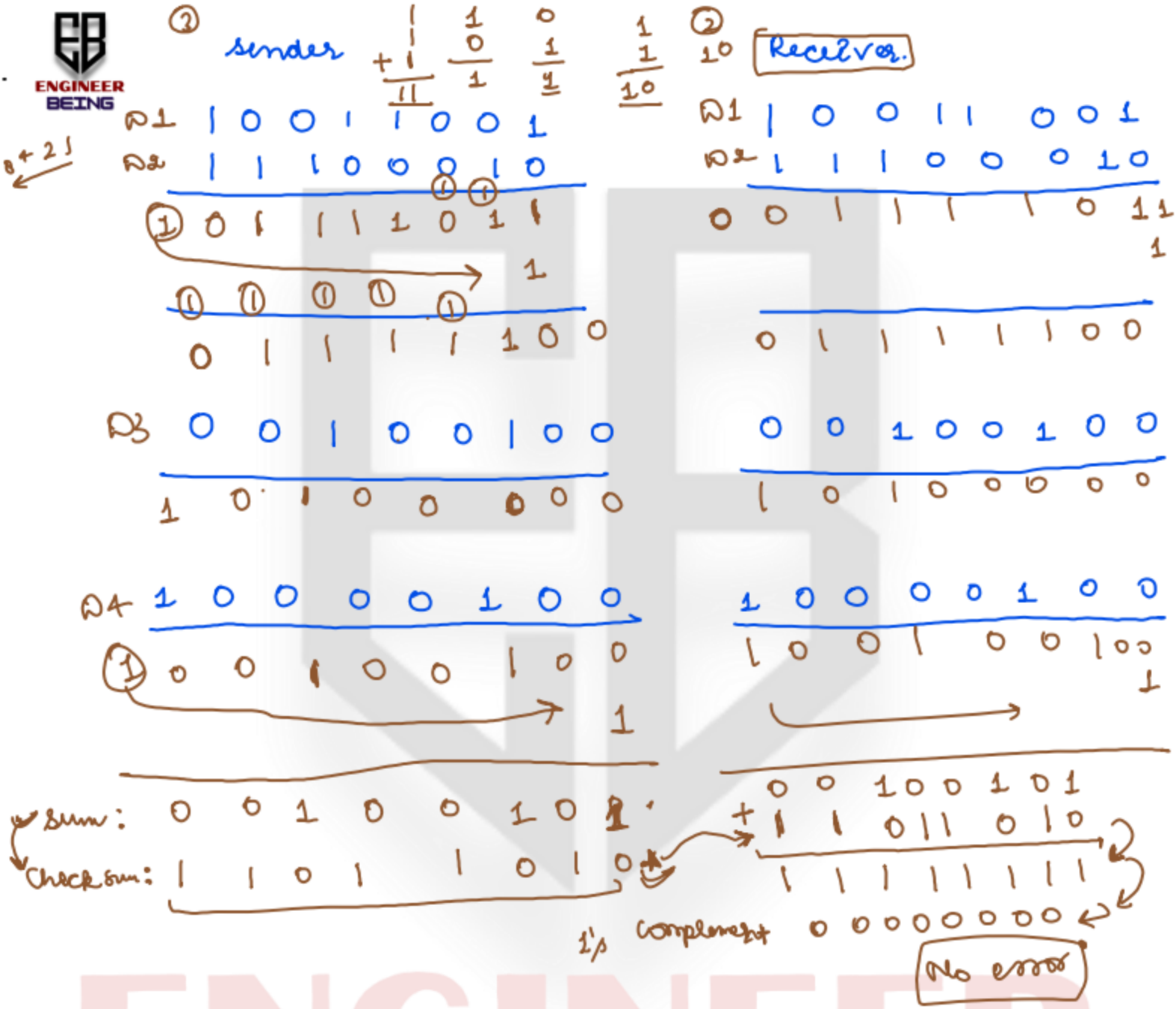
3. CHECKSUM

- In the checksum error detection scheme, the data is divided into k segments each of m bits.
- In the sender's end the segments are added using 1's complement arithmetic to get the sum.
- The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.

Original Data



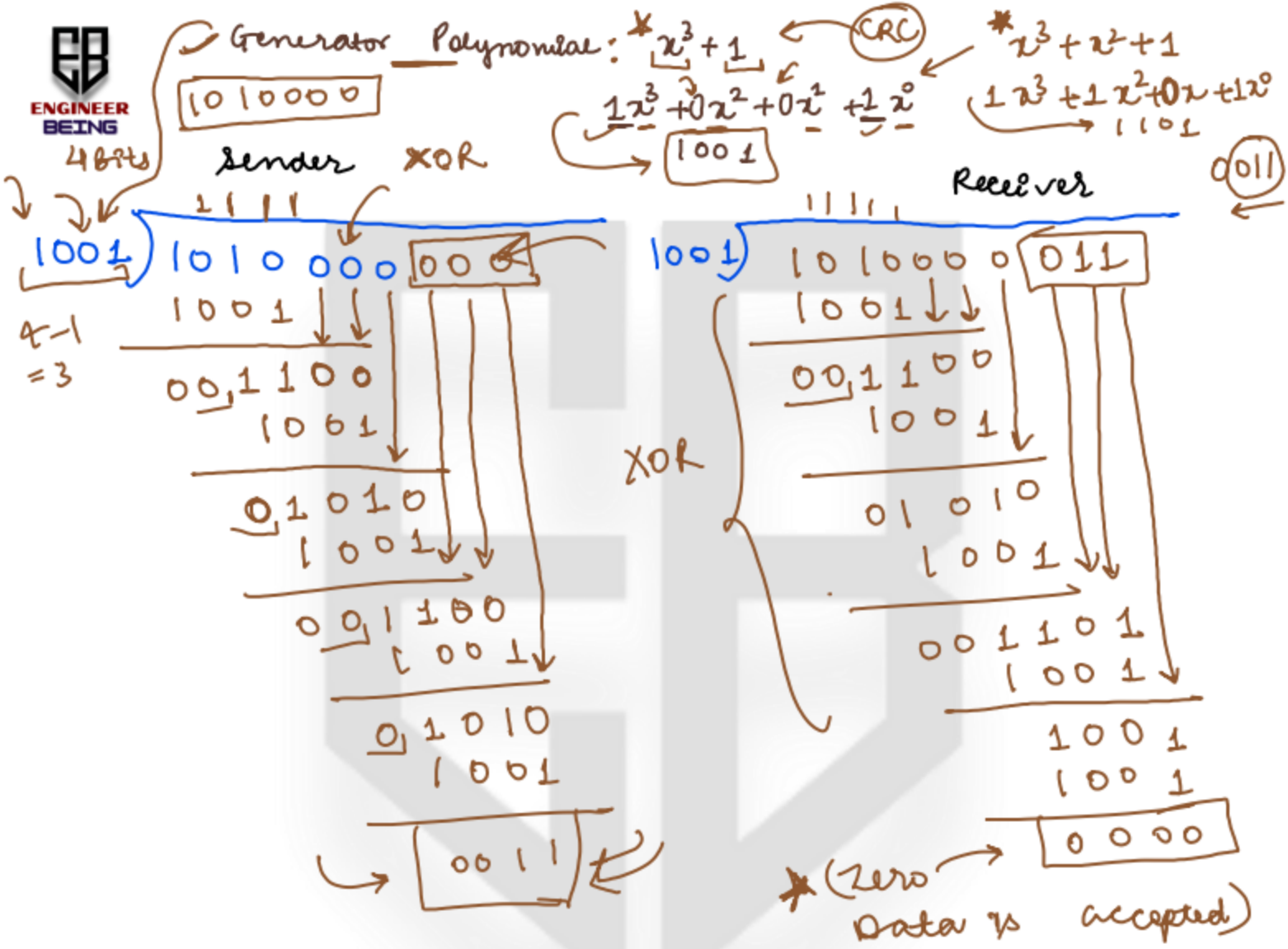
I. k=4 m= 8 Bits



4. CYCLIC REDUNDANCY CHECK (CRC)

- Unlike the checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number.
- If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

XOR →



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ERROR CORRECTION TECHNIQUE

- HAMMING CODES: ERROR CORRECTING CODES
- Tell error and position of error both.
- Single bit error correcting codes

Senders side

Ex= 1 0 1 0
a_3 a_2 a_1 a_0

1. Considered the data word $a_3\ a_2\ a_1\ a_0$

I. Calculate Redundant Bits (CHK)

$r_0 = (a_2 + a_1 + a_0) \bmod 2$ → $r_0 = (0+1+0) \% 2 = 1$
$r_1 = (a_3 + a_2 + a_1) \bmod 2$ → $r_1 = 0$
$r_2 = (a_3 + a_1 + a_0) \bmod 2$ → $r_2 = (1+1+0) \% 2 = 0$

Calculated codeword: $a_3\ a_2\ a_1\ a_0\ r_2\ r_1\ r_0$ ⇒ 1 0 1 0 0 0 1 *

$b_3\ b_2\ b_1\ b_0\ q_2\ q_1\ q_0$
1 0 1 1 0 0 1

Receiver side $a \Leftrightarrow b$, $r \Leftrightarrow q$

$b_3\ b_2\ b_1\ b_0\ q_2\ q_1\ q_0$
1 0 1 0 0 0 1

Calculate Syndrome Bits

$s_0 = (b_2 + b_1 + b_0 + q_0) \bmod 2$ → $s_0 = (0+1+1+1) \% 2 = 1$
$s_1 = (b_3 + b_2 + b_1 + q_1) \bmod 2$ → $s_1 = (1+0+1+0) \% 2 = 0$
$s_2 = (b_3 + b_1 + b_0 + q_2) \bmod 2$ → $s_2 = 1$

No error
0 0 0

$s_0 = (0+1+0+1) \% 2 = 0$
$s_1 = (0+1+1+0) \% 2 = 0$
$s_2 = (1+1+0+0) \% 2 = 0$

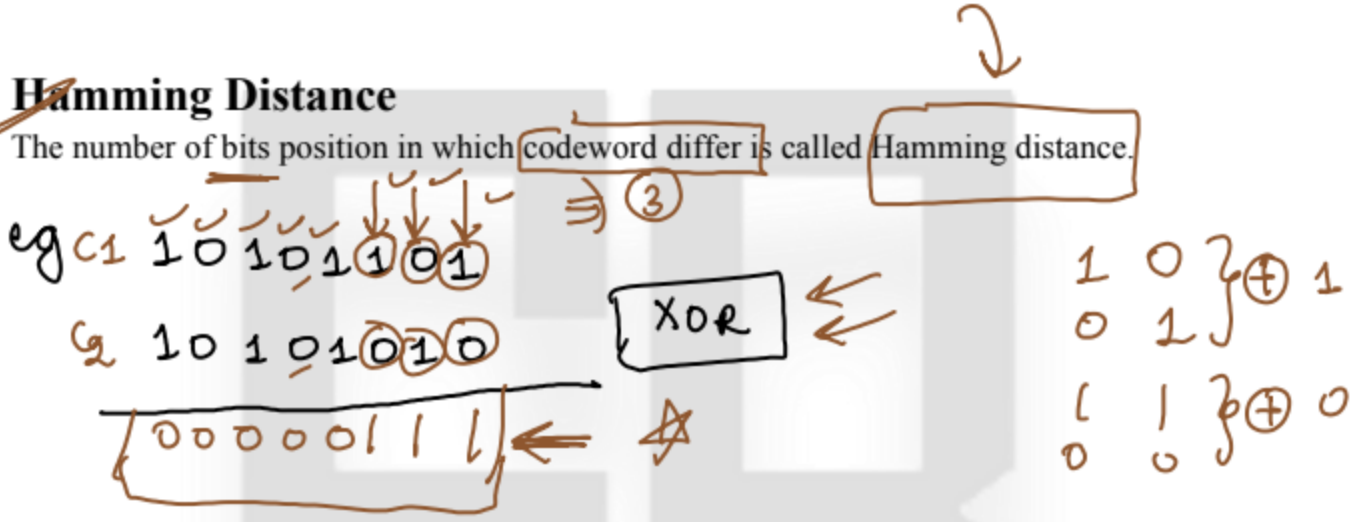
$s_2\ s_1\ s_0$
1 0 1 ← error

If all syndrome bits are zero then No error, else $s_2\ s_1\ s_0$ Will tell you the position in the data word where error lies.

Syndrome	000	001	010	011	100	101	110	111
Error	none	q_0	q_1	b_2	a_2	b_0	b_3	b_1

Hamming Distance

The number of bits position in which codeword differ is called Hamming distance.

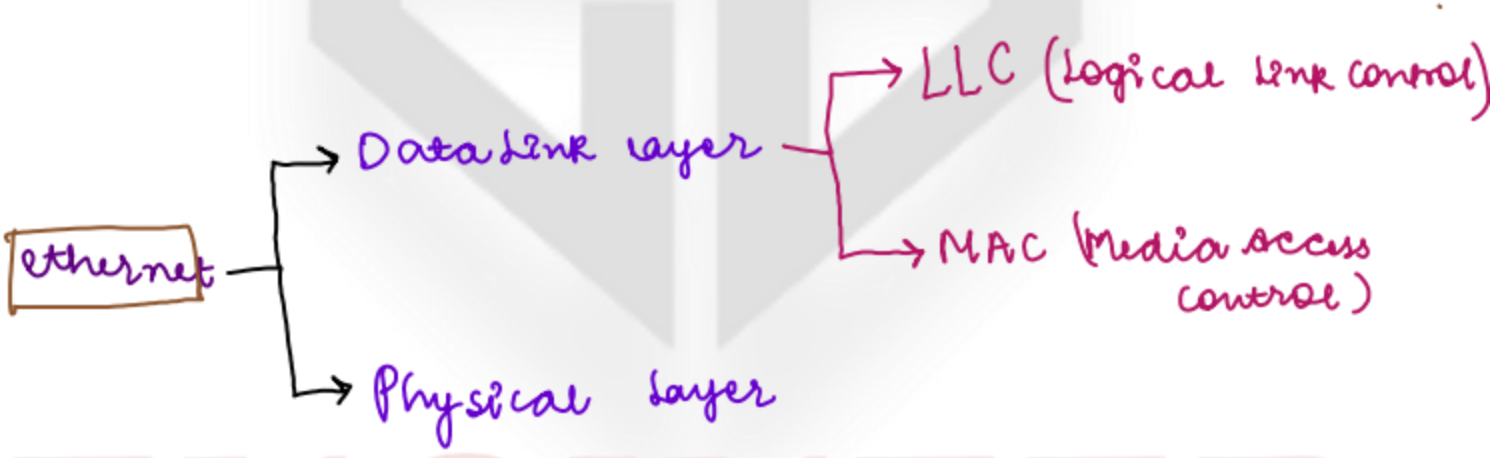


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LAN STANDARDS

ETHERNET

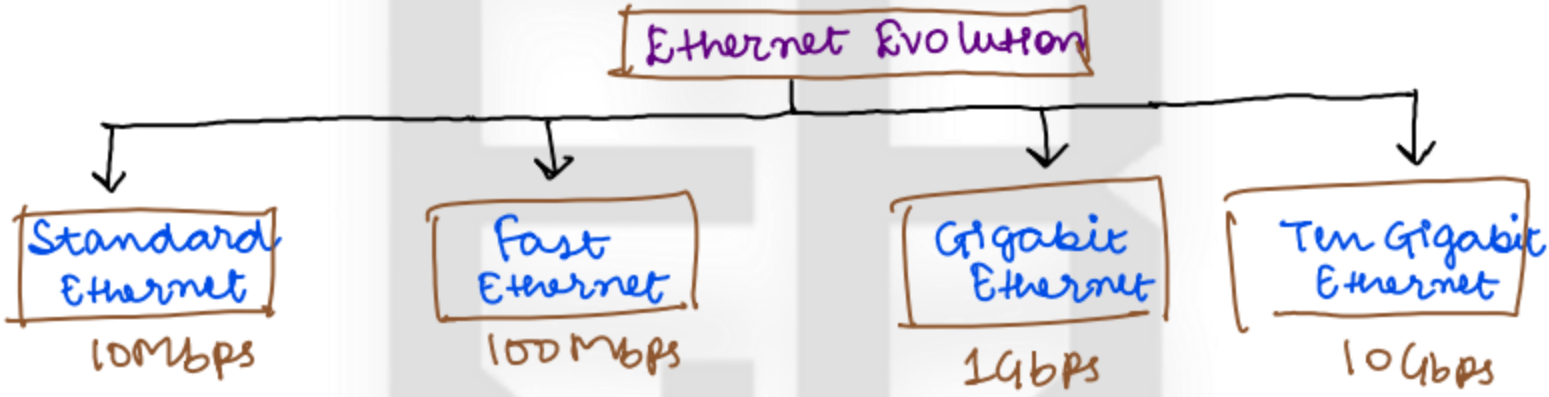
- Ethernet is a technology used to connect devices in a local area network (LAN).
- Ethernet is a wired communication technology that uses a physical connection, such as twisted-pair copper or fiber optic cable, to transmit data between devices in a local area network (LAN).
- The IEEE standard for Ethernet is known as IEEE 802.3. This standard defines the specifications for the Physical Layer and Data Link Layer protocols used in Ethernet networks.
- Access Method: CSMA/CD
  Standard Ethernet uses 1-persistent CSMA/CD
- Manchester Encoding



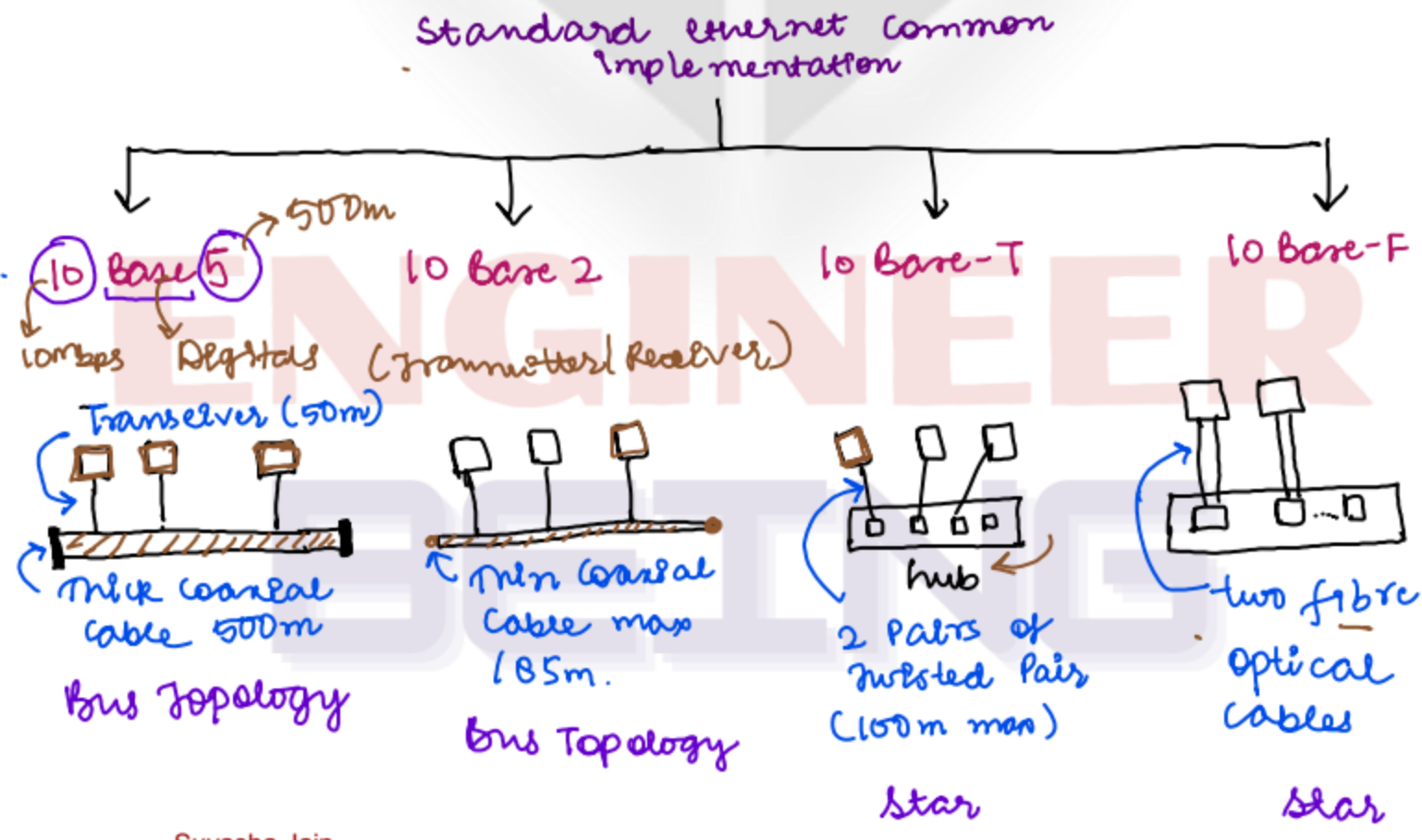
- The Physical Layer, also known as Layer 1, is responsible for transmitting the data over the physical medium, such as copper or fiber optic cable, and managing the electrical or optical signals used to transmit the data. The Physical Layer ensures that the data is transmitted over the cable in a reliable and efficient manner, and it defines the characteristics of the transmission medium, such as the cable type, transmission speed, and signaling method.
- DLL is sub divided into LLC AND MAC.

- The Data Link Layer, also known as Layer 2, is responsible for organizing the data into packets and adding addressing information. It ensures that the data packets are delivered to the correct device on the network by adding a unique identifier called the Media Access Control (MAC) address to each packet.

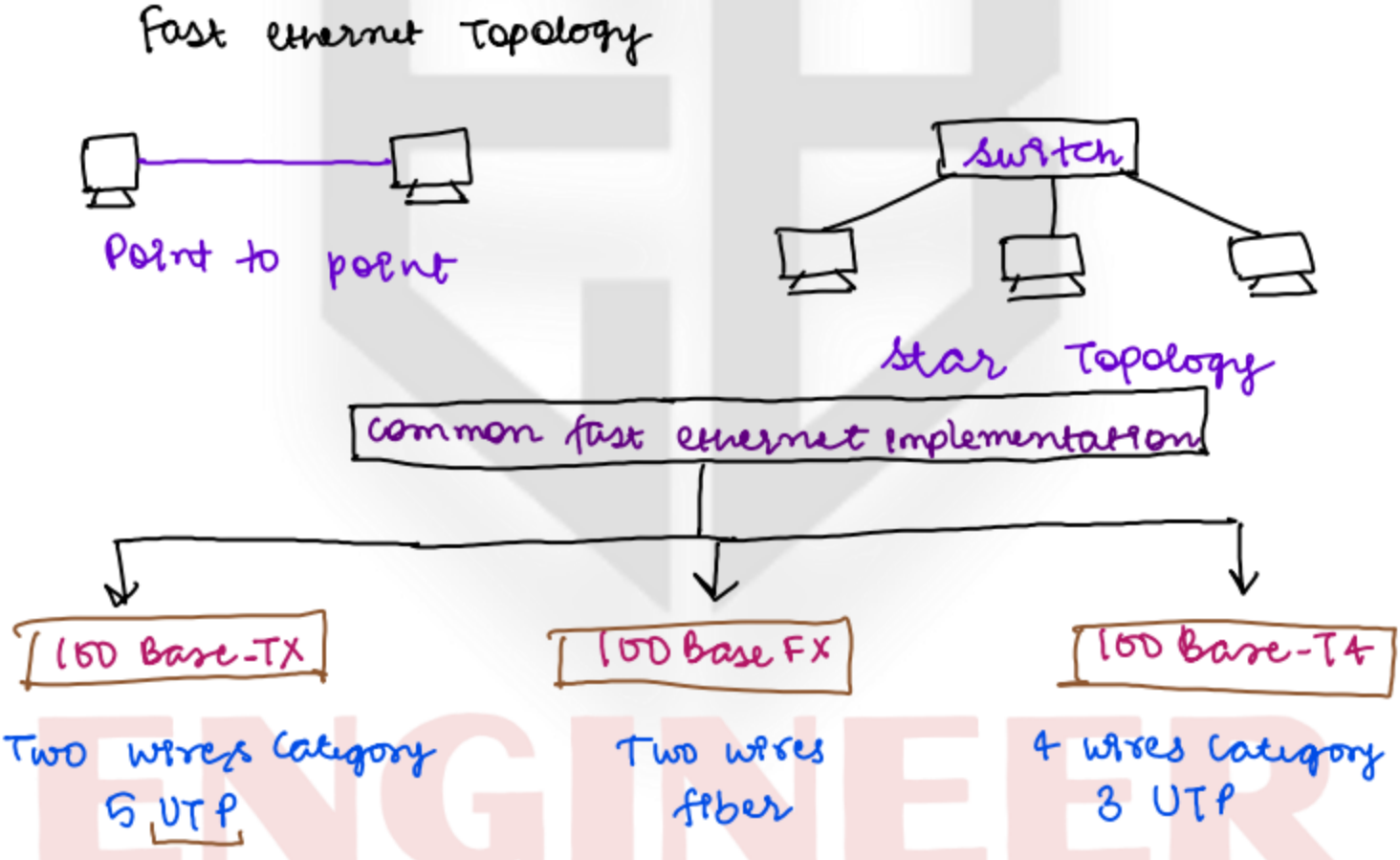
EVOLUTION OF ETHERNET



Standard Ethernet: A standard Ethernet network can transmit data at a rate up to 10 Megabits per second (10 Mbps)

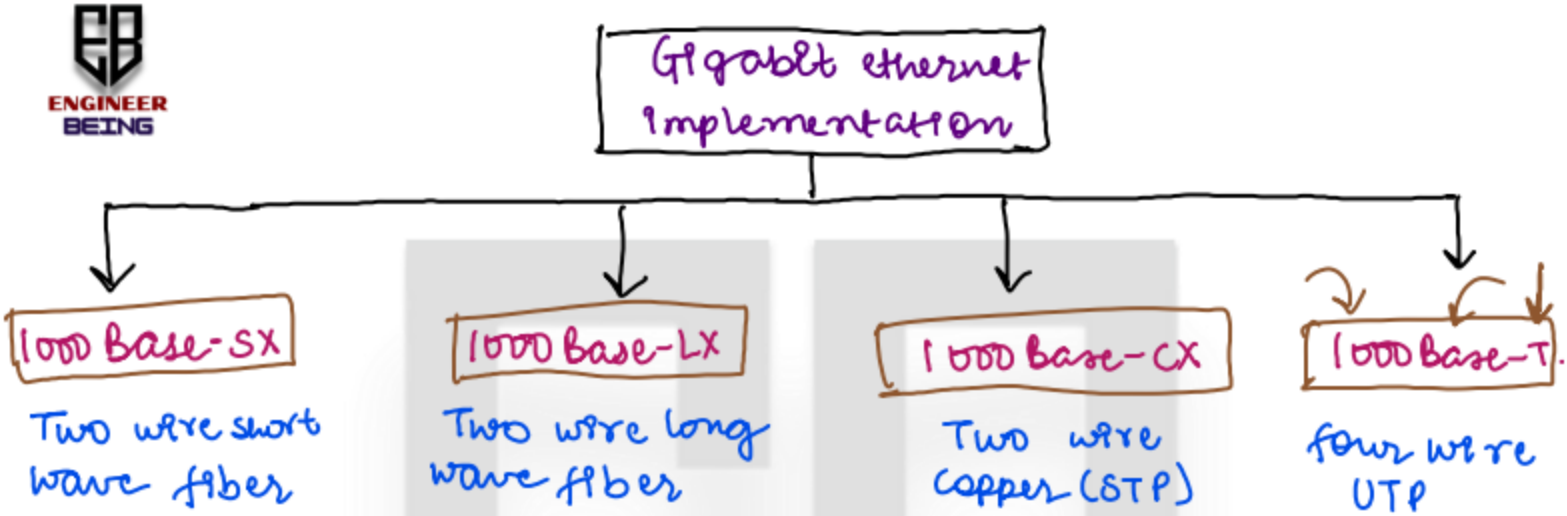


Fast Ethernet: This type of Ethernet is usually supported by a twisted pair or CAT5 cable, which has the potential to transfer or receive data at around 100 Mbps. They function at 100Base and 10/100Base Ethernet on the fiber side of the link if any device such as a camera, laptop, or other is connected to a network. The fiber optic cable and twisted pair cable are used by fast Ethernet to create communication. The 100BASE-TX, 100BASE-FX, and 100BASE-T4 are the three categories of Fast Ethernet.

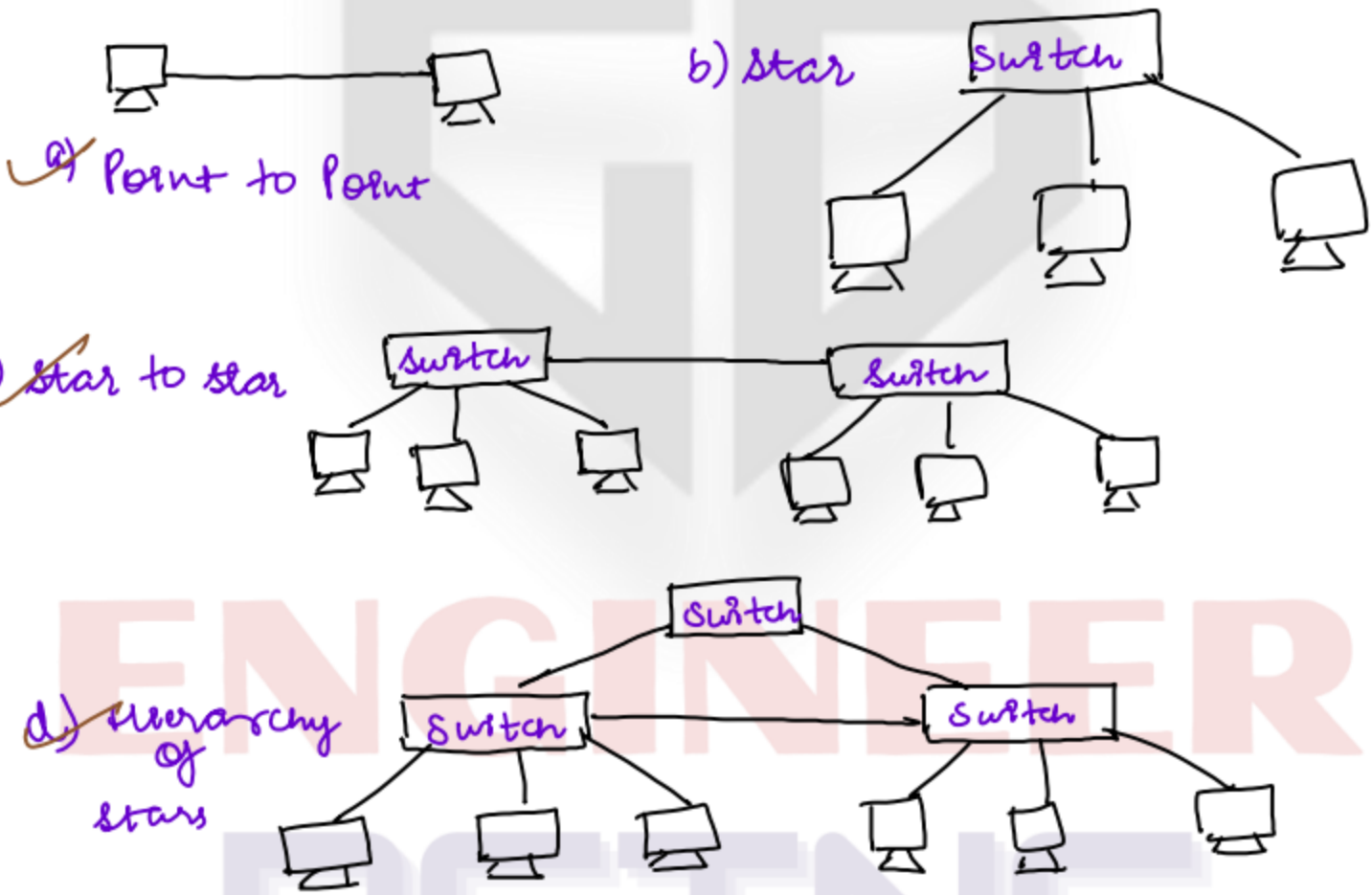


Gigabit Ethernet: This type of Ethernet network is an upgrade from Fast Ethernet, which uses fiber optic cable and twisted pair cable to create communication. It can transfer data at a rate of 1000 Mbps or 1Gbps. In modern times, gigabit Ethernet is more common. This network type also uses CAT5e or other advanced cables, which can transfer data at a rate of 10 Gbps.The primary intention of developing the gigabit Ethernet was to full fill the user's requirements, such as faster transfer of data, faster communication network, and more.

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Gigabit ethernet topology

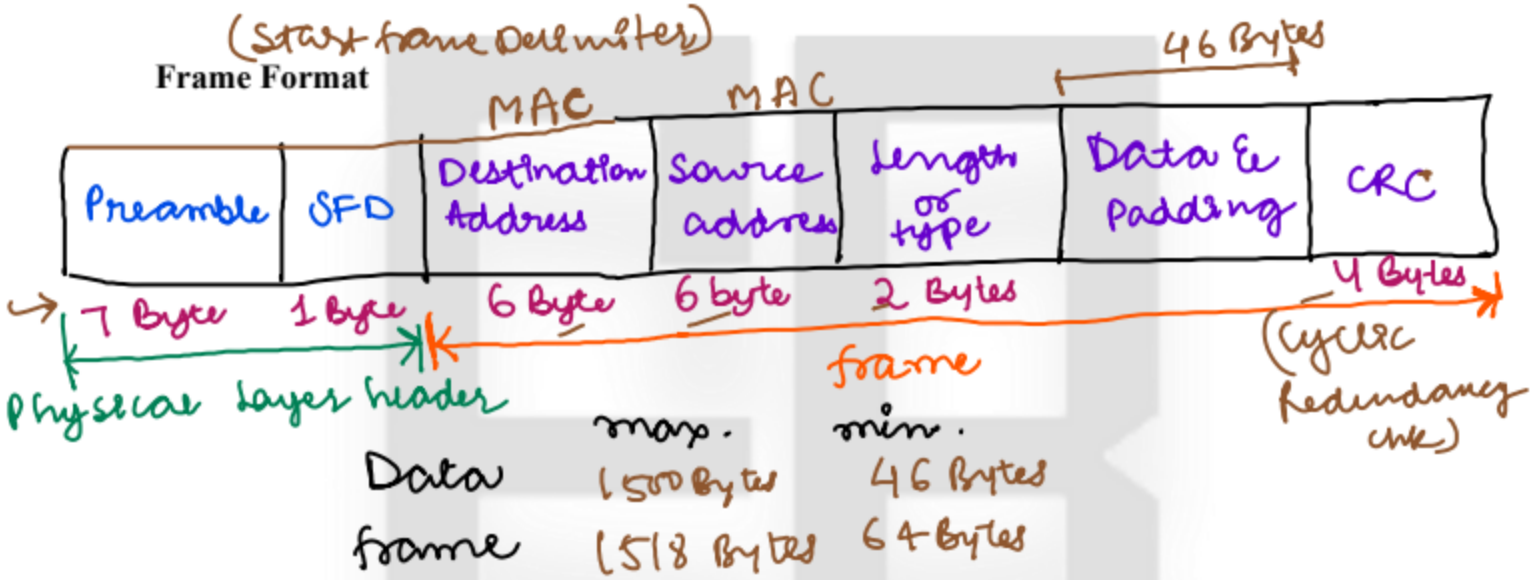


10-Gigabit Ethernet: This type of network can transmit data at a rate of 10 Gigabit/second, considered a more advanced and high-speed network. It makes use of CAT6a or CAT7 twisted-pair cables and fiber optic cables as well. This network can be expanded up to nearly 10,000 meters with the help of using a fiber optic cable.

Preamble – 56 Bits of alternating 10101010 ---

SFD – Start frame Delimeter 10101011

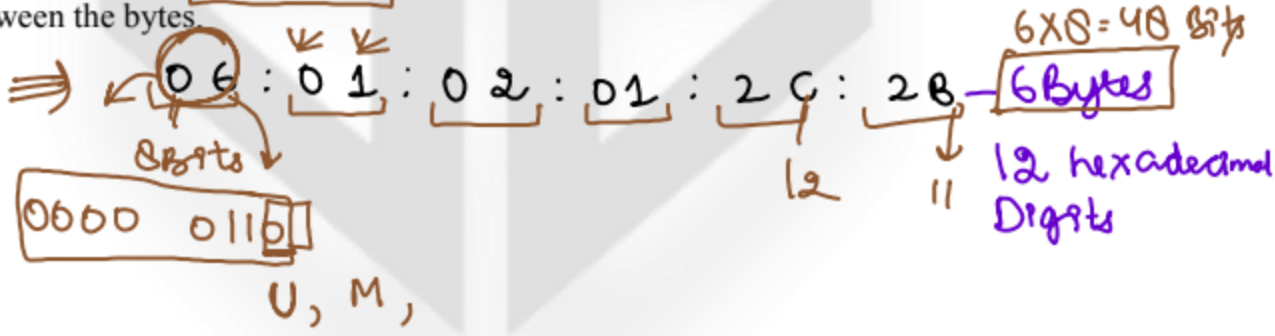
Frame Format



	max.	min.
Data	1500 Bytes	46 Bytes
frame	1518 Bytes	64 Bytes

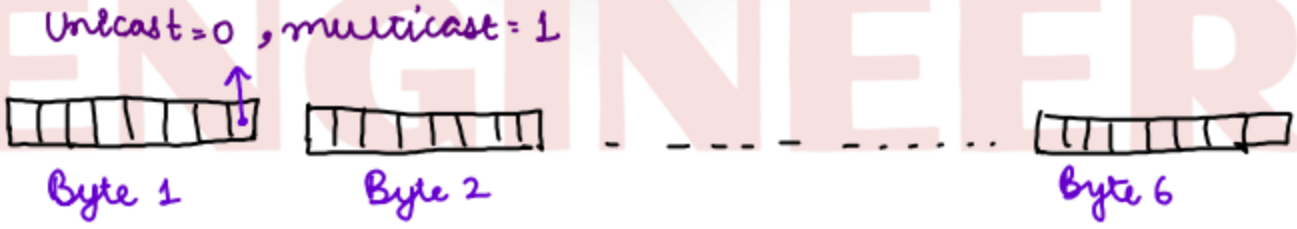
The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes.

Example



A-10
B-11
C-12
D-13
E-14
F-15

Unicast, Multicast, and Broadcast Addresses



If the least significant bit of the first byte in a destination address is 0, the address is unicast; otherwise, it is multicast.
A broadcast destination address has all forty-eight bits as 1.

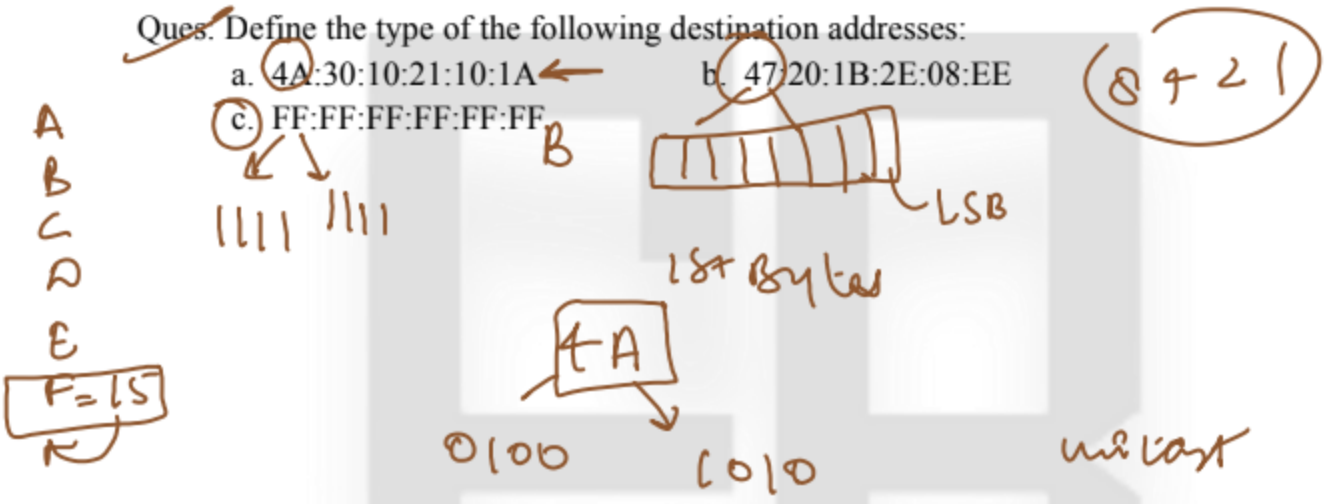
A source address is always a unicast address-the frame comes from only one station.
The destination address, however, can be unicast, multicast, or broadcast.

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Ques. Define the type of the following destination addresses:

a. 4A:30:10:21:10:1A ← b. 47:20:1B:2E:08:EE

c. FF:FF:FF:FF:FF:FF



FDDI (Fiber Distributed Data Interface)

FDDI (Fiber Distributed Data Interface) is a LAN (Local Area Network) standard that was developed in the 1980s as an alternative to Ethernet. FDDI is a high-speed, fiber optic network technology that is capable of transmitting data at up to 100 Mbps over distances of up to 200 kilometers. FDDI networks are typically used in large enterprise environments that require high bandwidth and reliability, such as universities, government agencies, and financial institutions. However, FDDI has largely been superseded by newer network technologies such as Gigabit Ethernet and 10 Gigabit Ethernet, which offer even higher speeds and greater flexibility at lower costs.

Token Ring

Token Ring is a LAN (Local Area Network) standard that was developed in the 1980s as an alternative to Ethernet. Token Ring networks use a token passing protocol to control access to the network and prevent collisions between data packets. In a Token Ring network, a token is passed around the network, and only the device that holds the token is allowed to transmit data. This approach helps to ensure that the network bandwidth is used efficiently and that data is transmitted reliably. Differential Manchester is used for encoding. (Data Rate: 4 to 18Mbps)

Token Ring networks were initially popular in certain industries such as banking and finance, where reliability and security were critical requirements. However, Token Ring networks have largely been replaced by Ethernet and other LAN technologies due to their higher costs, slower speeds, and limited scalability compared to Ethernet.

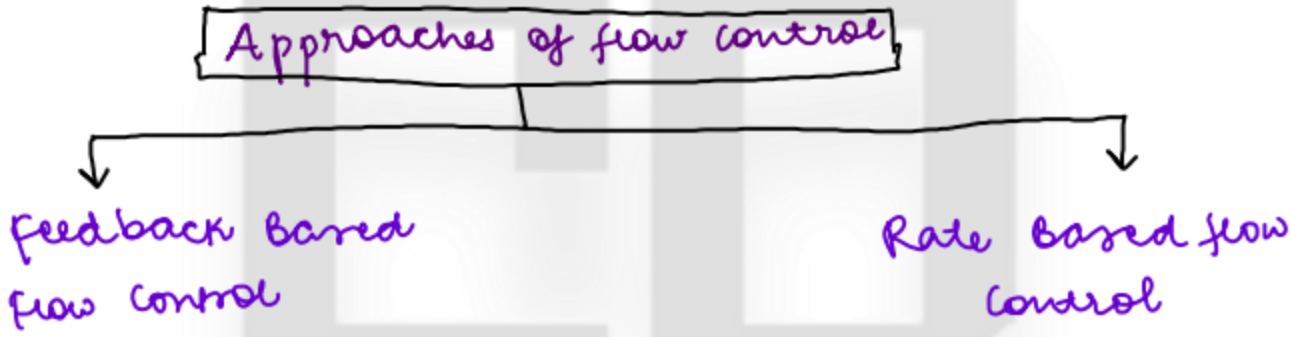
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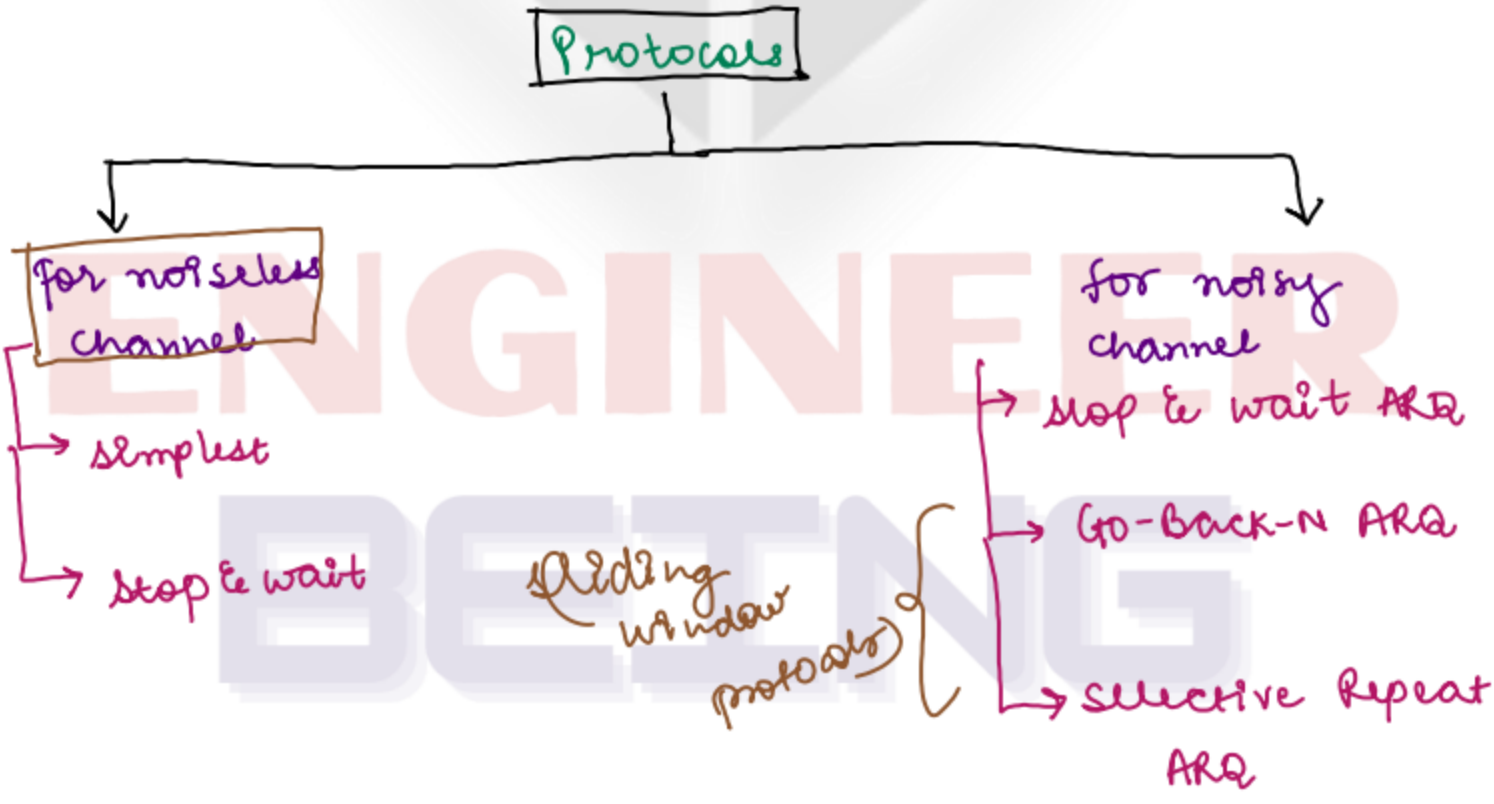
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FLOW CONTROL



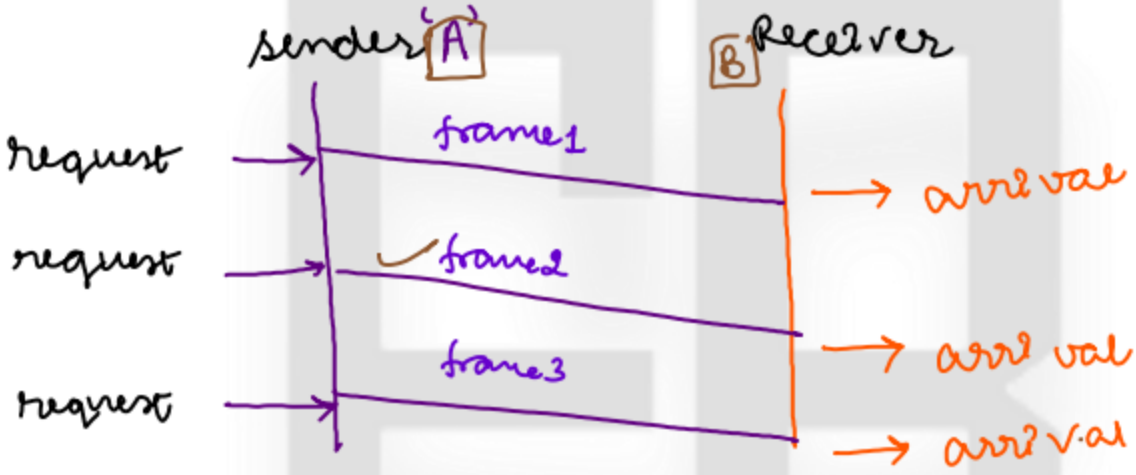
Feedback based Flow Control: In these protocols, the sender sends frames after it has received acknowledgments from the user. This is used in the data link layer.

Rate based Flow Control: These protocols have built in mechanisms to restrict the rate of transmission of data without requiring acknowledgment from the receiver. This is used in the network layer and the transport layer.



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SIMPLEST PROTOCOL



The following assumptions have been made for developing the simplest protocol:

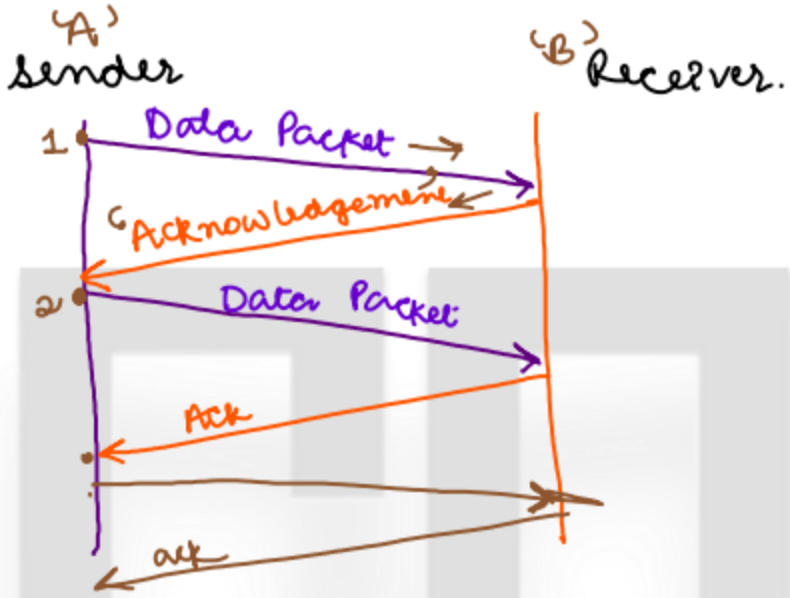
- The transmission channel is completely noiseless (a channel in which no frames are lost, corrupted, or duplicated).
- The transmission channel is assumed to be ideal in which there is no data loss, no data duplication, and no data corruption.
- There is no error and flow control mechanism.
- The buffer space for storing the frames at the sender's end and the receiver's end is infinite.

We assume that the receiver is always ready to handle any frames coming from the sender immediately. The simplest protocol is a unidirectional protocol in which the data frames are traveling in only one direction from the sender to the receiver.

Since the simplest protocol is unidirectional, there is no acknowledgment (ACK). Also, as there is no data loss in the transmission, there is no need for data retransmission.

2 STOP AND WAIT PROTOCOL

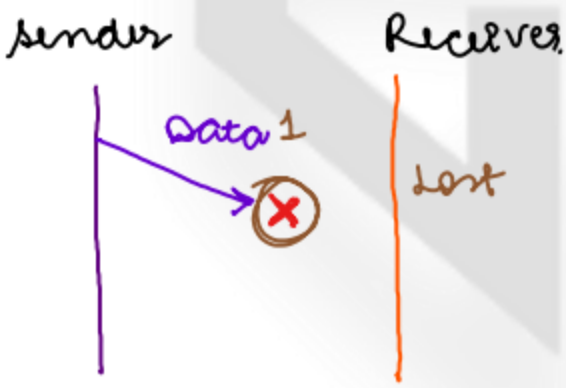
It is still very simple. The sender sends one frame and waits for feedback from the receiver. When the ACK arrives, the sender sends the next frame. It is Stop-and-Wait Protocol because the sender sends one frame, stops until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame.



DISADVANTAGE:

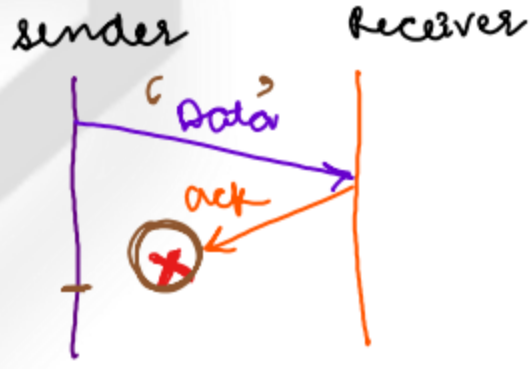
1. Problems occur due to lost data
2. Problems occur due to lost acknowledgment
3. Problem due to delayed Ack/Data

1)



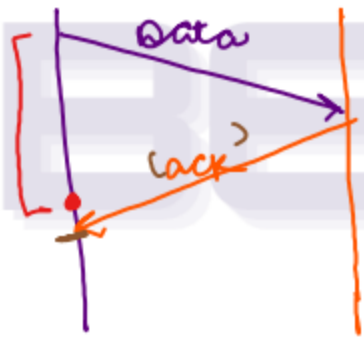
Sender wait for ack for
a time & Receiver wait
for Data for a time

2



Sender wait for a time
for ack.

3)



"Delayed ack will not be considered
Wrong"

"Time counting"

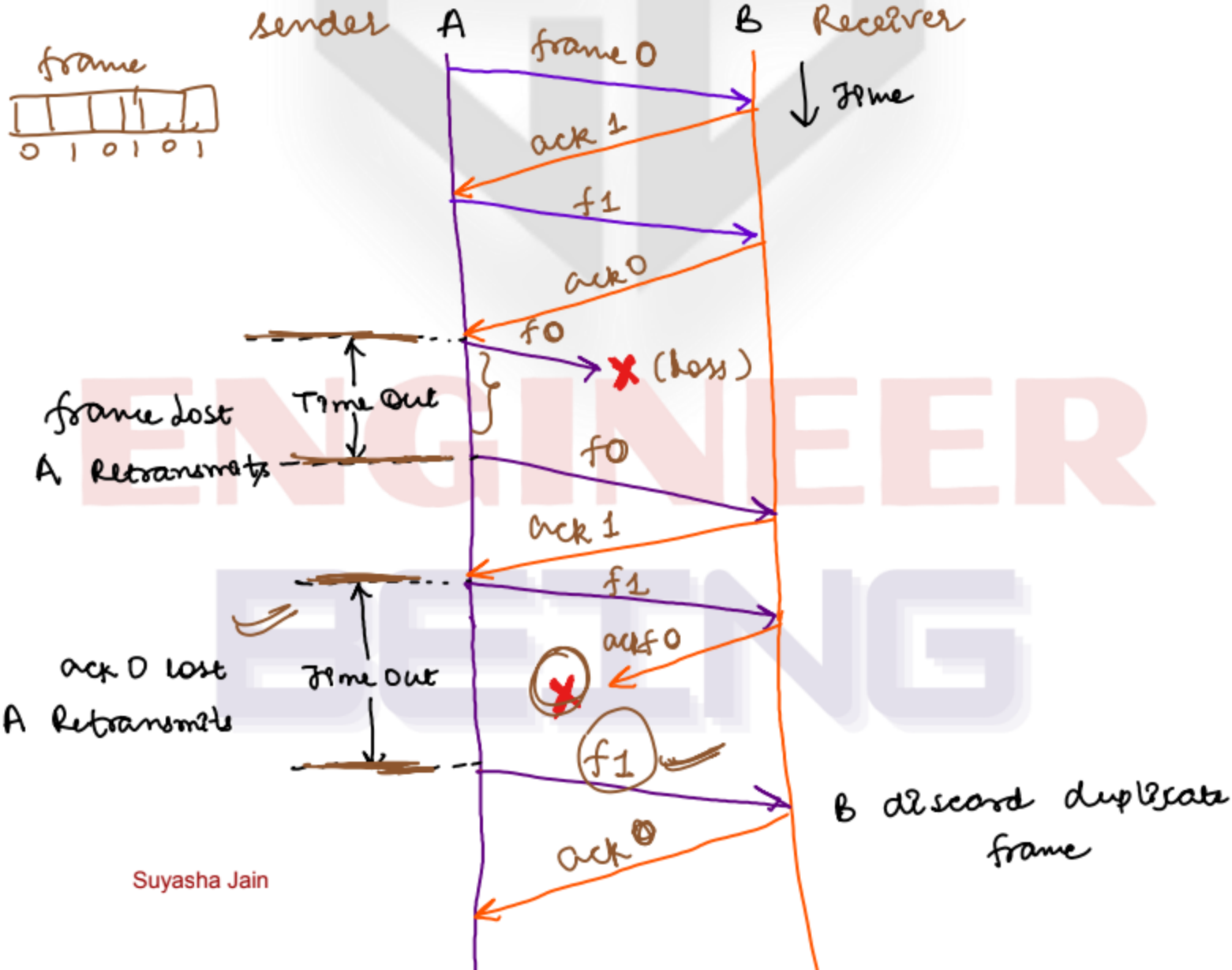
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STOP AND WAIT ARQ(Automatic Repeat Request)

- Used in Connection-oriented communication.
- It offers error and flows control
- It is used in Data Link and Transport Layers
- Stop and Wait for ARQ mainly implements the Sliding Window Protocol concept with Window Size 1

Working

1) Sender A sends a data frame or packet with sequence number 0.
2) Receiver B, after receiving the data frame, sends an acknowledgement with sequence number 1 (the sequence number of the next expected data frame or packet)
There is only a one-bit sequence number that implies that both sender and receiver have a buffer for one frame or packet only.



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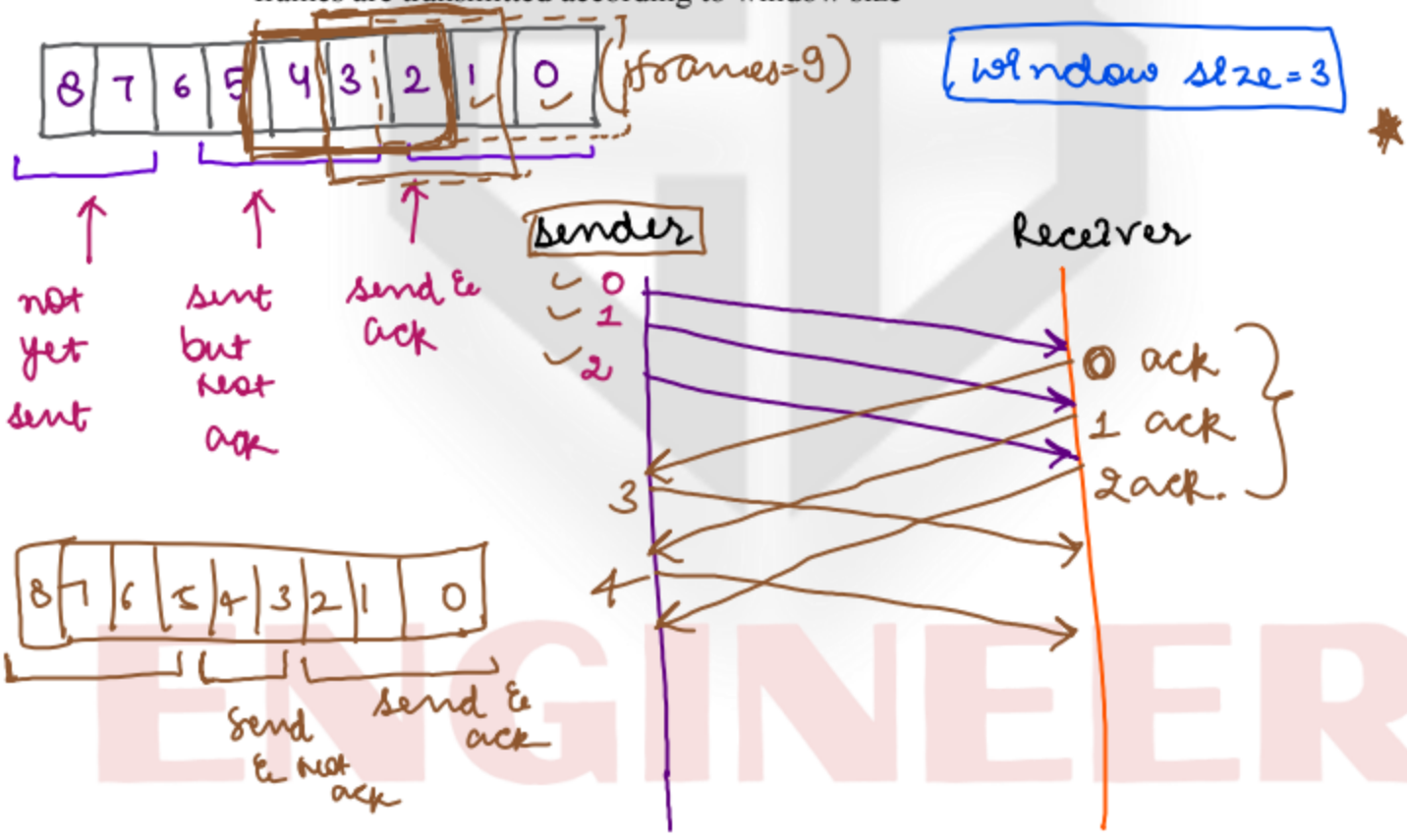
DISADVANTAGE:

- One frame transmission at time
- Poor utilization of bandwidth

SLIDING WINDOW PROTOCOL

Noisy Channel.

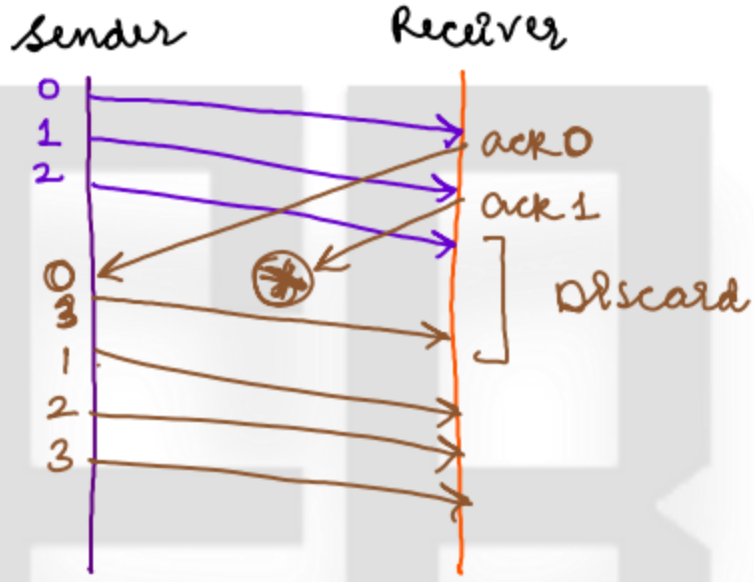
- technique for sending multiple frames at a time.
- It is also used in TCP (Transmission Control Protocol).
- Each frame is numbered: sequence number
- frames are transmitted according to window size



GO-BACK-N ARQ(Automatic Repeat Request)

- If a frame is corrupted or lost in it, all subsequent frames have to be sent again.
- If it has a high error rate, it wastes a lot of bandwidth.
- Number of frames sent depends on window size.
- The size of the sender window is N in this protocol. For example, Go-Back-8, the size of the sender window, will be 8. The receiver window size is always 1



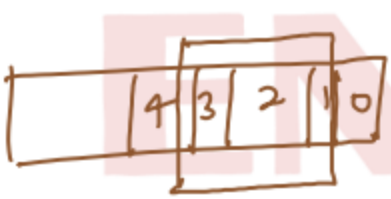


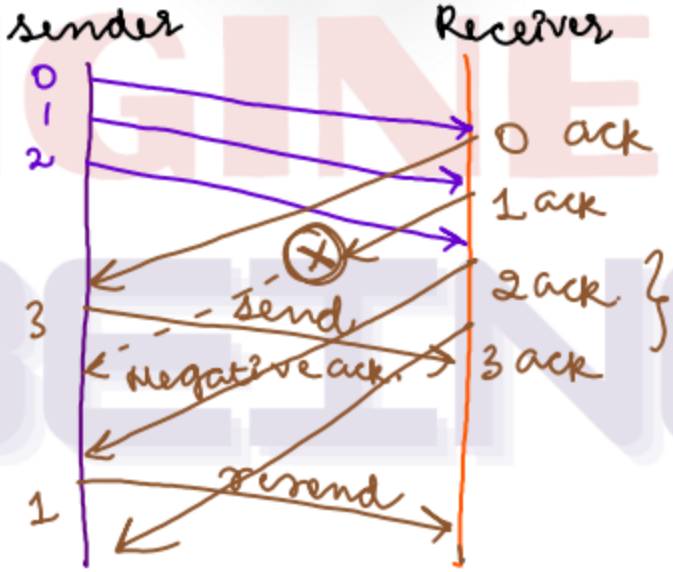
SELECTIVE REPEAT ARQ(Automatic Repeat Request)

- In this only the frame is sent again, which is lost.
- There is loss of low bandwidth
- In this protocol, the size of the sender window is always equal to the size of the receiver window. The size of the sliding window is always greater than 1.
- If the receiver receives a corrupt frame, it does not directly discard it. It sends a negative acknowledgement to the sender. The sender sends that frame again as soon as on the receiving negative acknowledgment.

(N, 1)





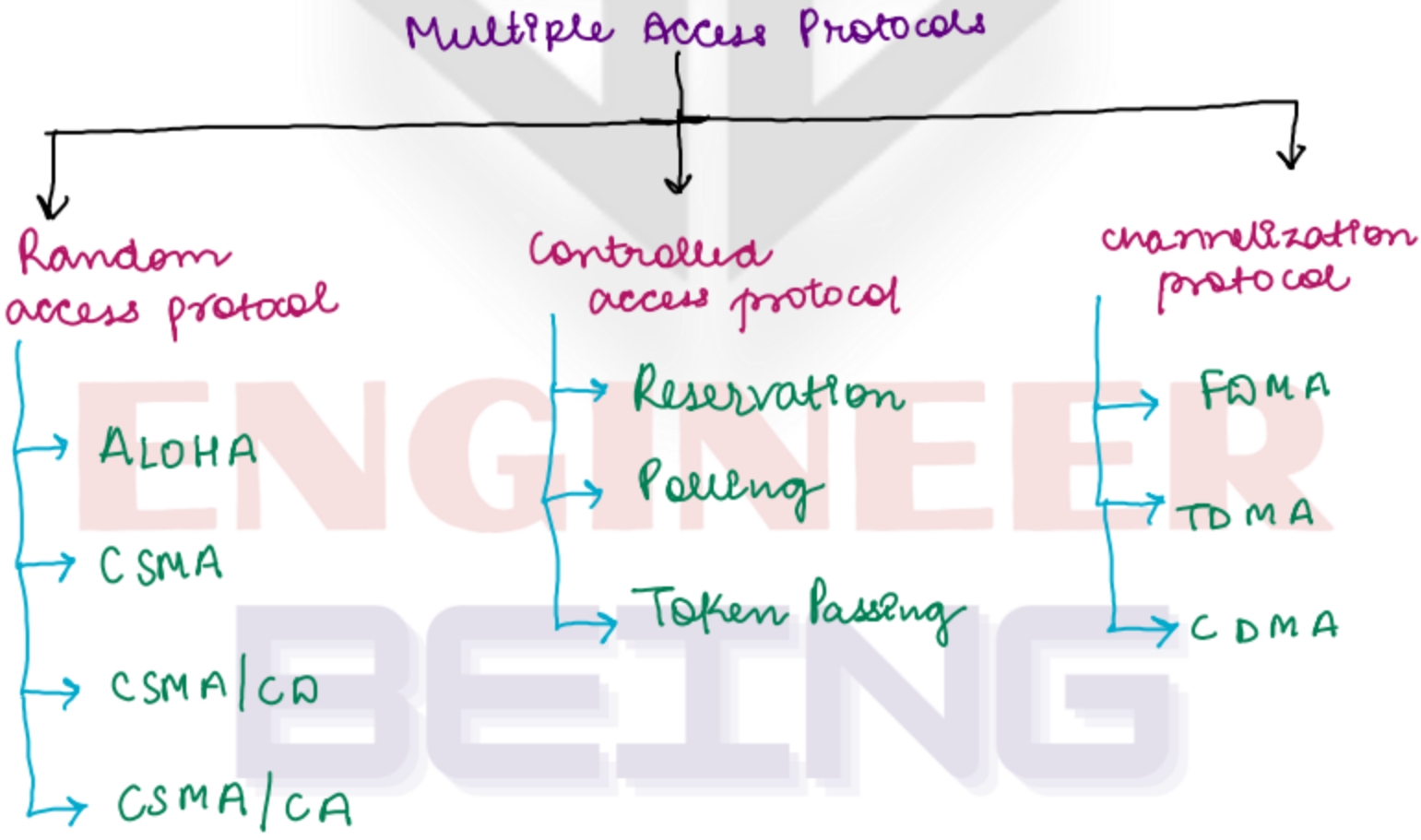


MULTIPLE ACCESS CONTROL PROTOCOL



The upper sub layer that is responsible for flow and error control is called the logical link control (LLC) layer

The lower sub layer that is mostly responsible for multiple access resolution is called the media access control (MAC) layer. When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.



ENGINEER BEING

Random Access Protocols

Random access protocols are communication protocols that allow multiple devices to access a shared communication channel or medium in random or arbitrary order. Devices access the medium by using a method called random access.
Random access protocols refer to communication protocols used in computer networks. Hence, each device has an equal privilege to access the network and transmit data. These protocols allow nodes to send and receive data without taking turns or waiting for permission.

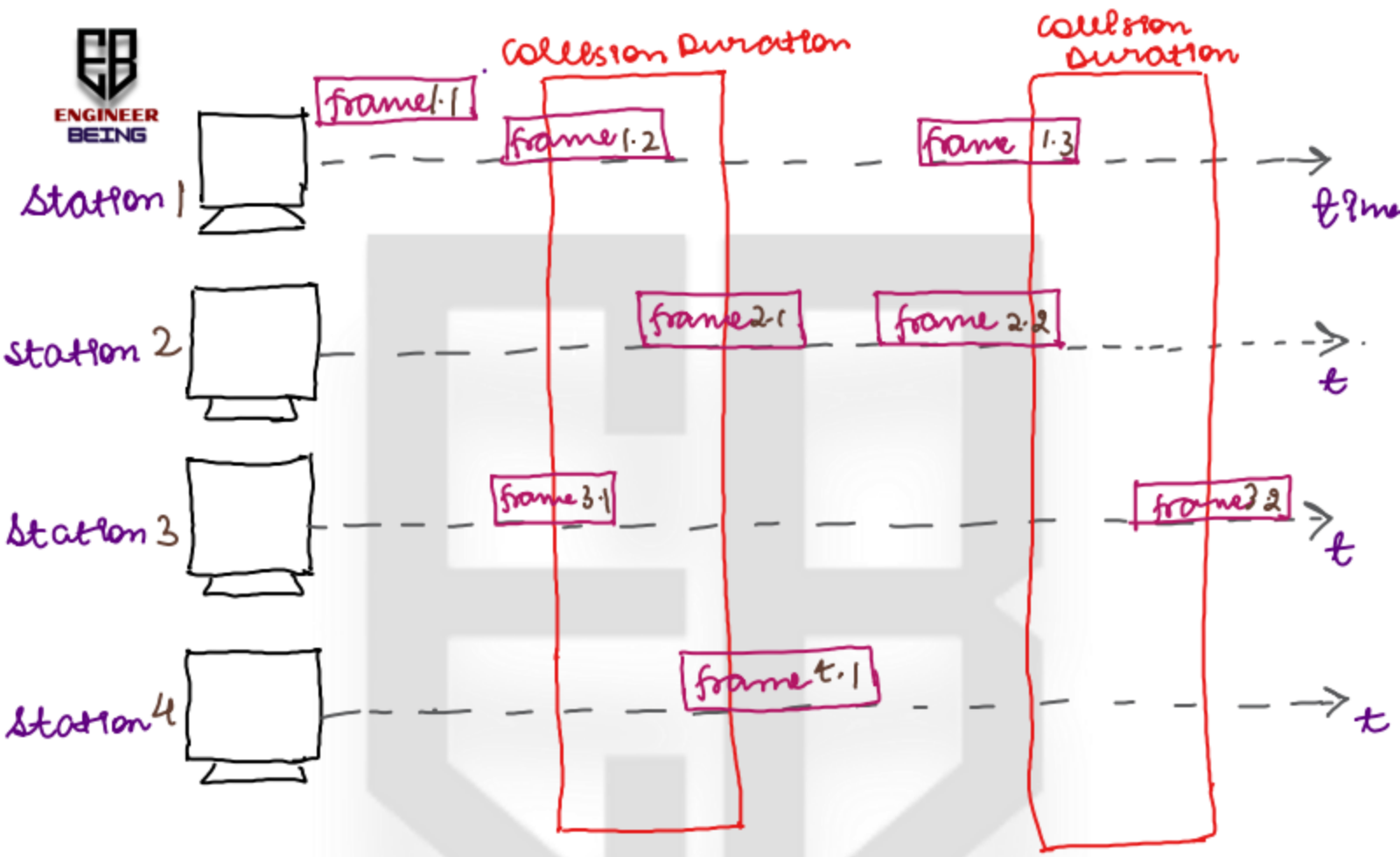
ALOHA Random Access Protocol

ALOHA is an early random access protocol used in computer networks.
Devices can transmit data whenever they have a message to send



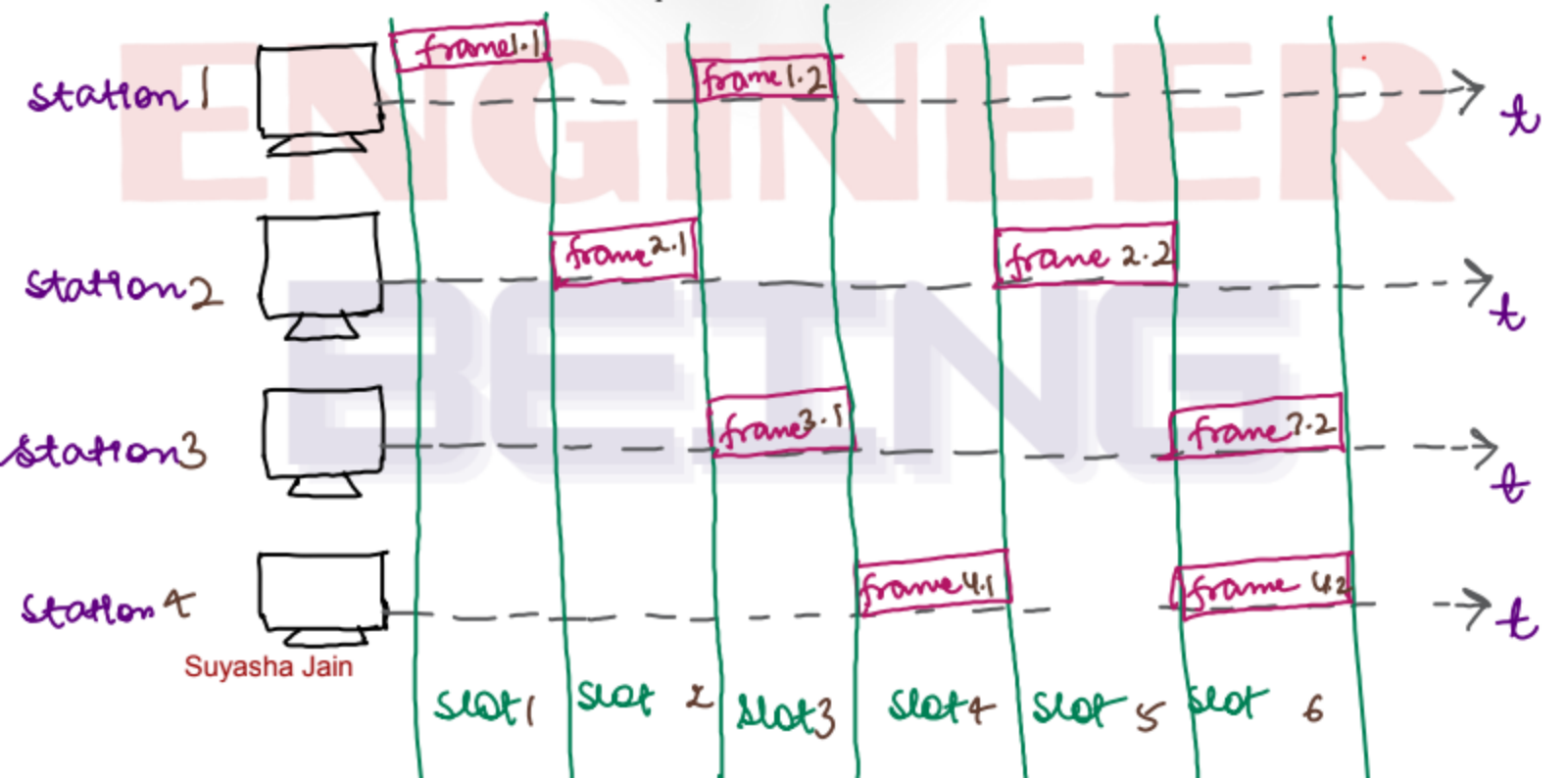
Pure ALOHA

- In this version, devices can transmit at any time. There is no coordination between devices.
- If two or more devices transmit simultaneously, their messages will collide and be corrupted
- Devices that detect a collision will wait for a random amount of time before trying to transmit again
- The total vulnerable time of pure Aloha is 2 * Tfr.
- The throughput for pure ALOHA is $S = G \times e^{-2G}$.
- The maximum throughput Smax = 0.184 when G= (1/2).



Slotted ALOHA

- The slotted ALOHA divides time into slots and only allows devices to transmit at the beginning of a slot, which reduces the probability of collision
- The throughput for slotted ALOHA is $S = G \times e^{-G}$. The maximum throughput $S_{max} = 0.368$ when $G = 1$.
- The total vulnerable time required in slotted Aloha is Tfr.



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CSMA (Carrier Sense Multiple Access)

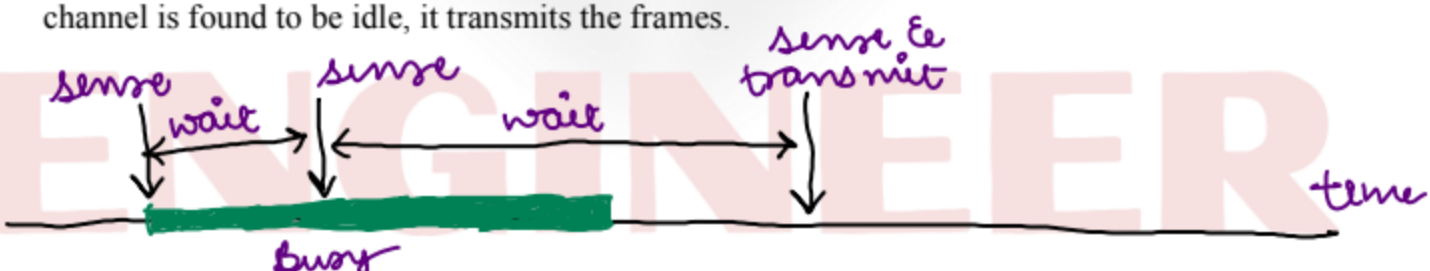
- It is a carrier sense multiple access based on media access protocol to sense the traffic on a channel (idle or busy) before transmitting the data.
- It means that if the channel is idle, the station can send data to the channel. Otherwise, it must wait until the channel becomes idle.
- Hence, it reduces the chances of a collision on a transmission medium.

CSMA Access Modes

I-Persistent: In the I-Persistent mode of CSMA that defines each node, first sense the shared channel and if the channel is idle, it immediately sends the data. Else it must wait and keep track of the status of the channel to be idle and broadcast the frame unconditionally as soon as the channel is idle.

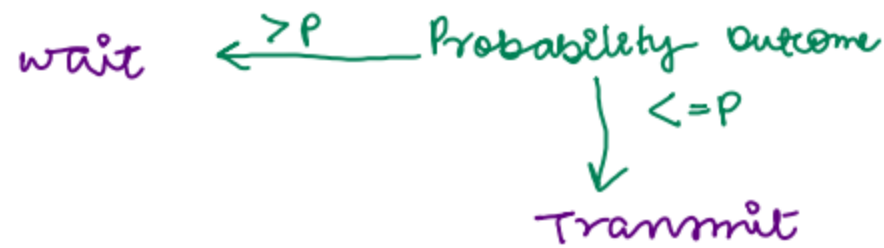


Non-Persistent: It is the access mode of CSMA that defines before transmitting the data, each node must sense the channel, and if the channel is inactive, it immediately sends the data. Otherwise, the station must wait for a random time (not continuously), and when the channel is found to be idle, it transmits the frames.



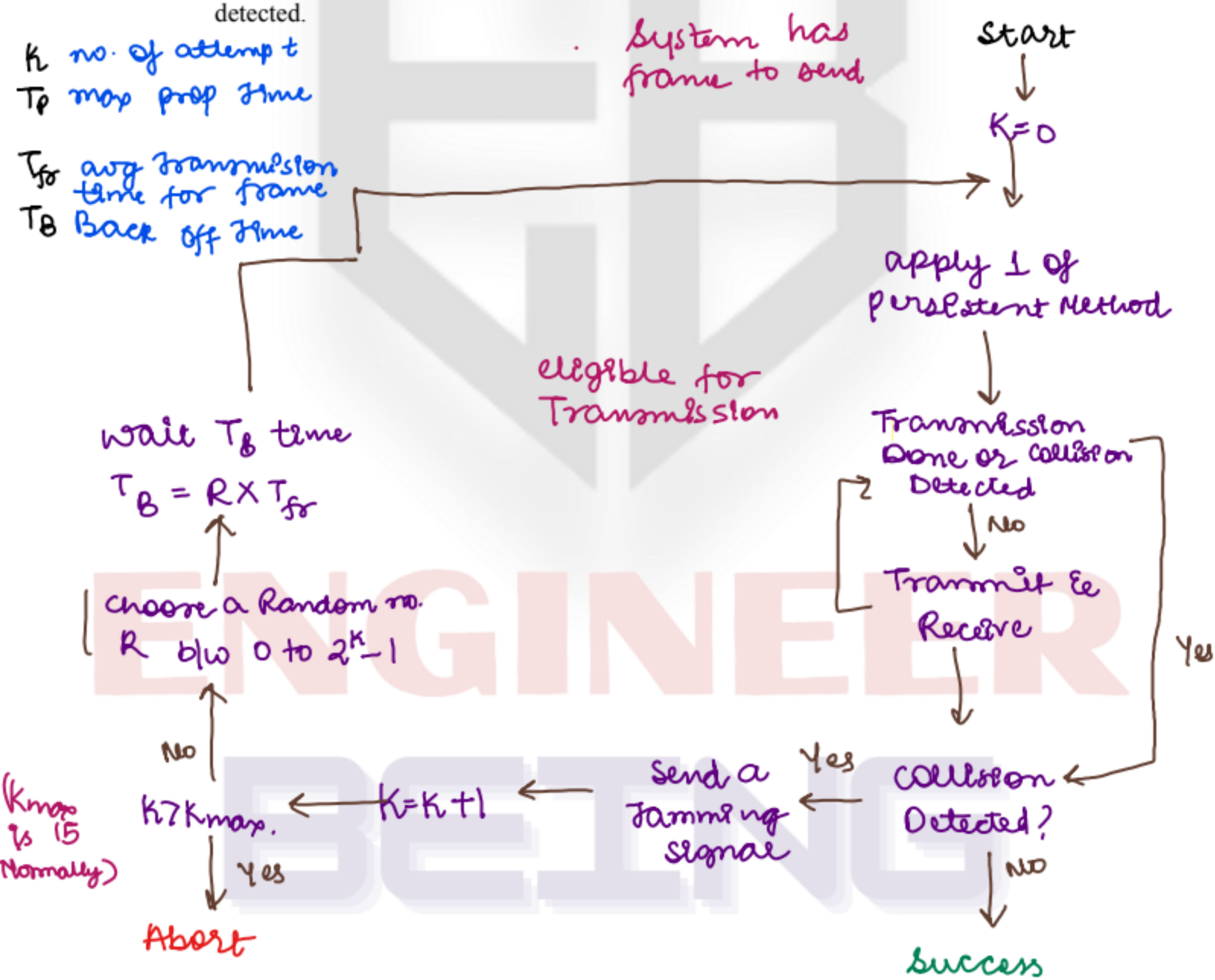
P-Persistent: It is the combination of 1-Persistent and Non-persistent modes. The P-Persistent mode defines that each node senses the channel, and if the channel is inactive, it sends a frame with a P probability. If the data is not transmitted, it waits for a (q = 1-p probability) random time and resumes the frame with the next time slot.





CSMA/CD

- While transmitting, the device continues to listen to the network to detect any collisions. If a collision is detected, the device will immediately stop transmitting and send a jam signal to indicate that a collision has occurred. The device will then wait for a random amount of time before trying to retransmit its data.
- The goal of collision detection is to minimize the impact of collisions on the network by ensuring that devices stop transmitting as soon as a collision is detected.



Three Strategies → IFS (Interframe space)
→ contention window
→ Acknowledgement

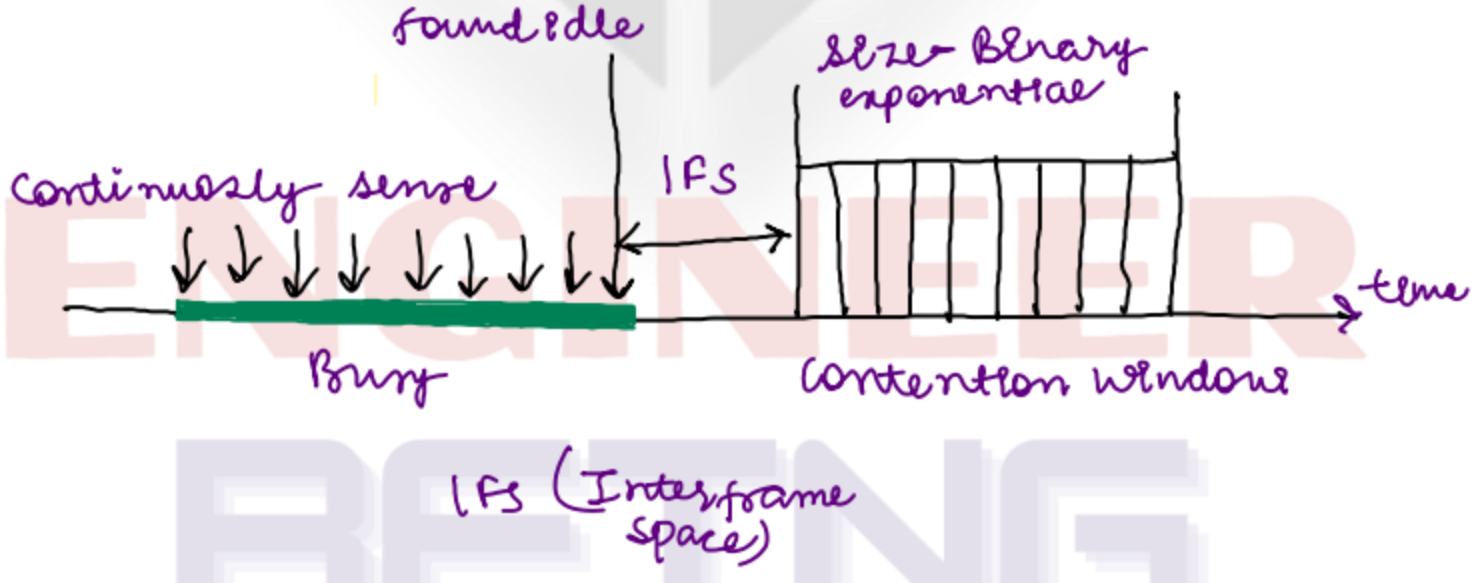
CSMA/CA

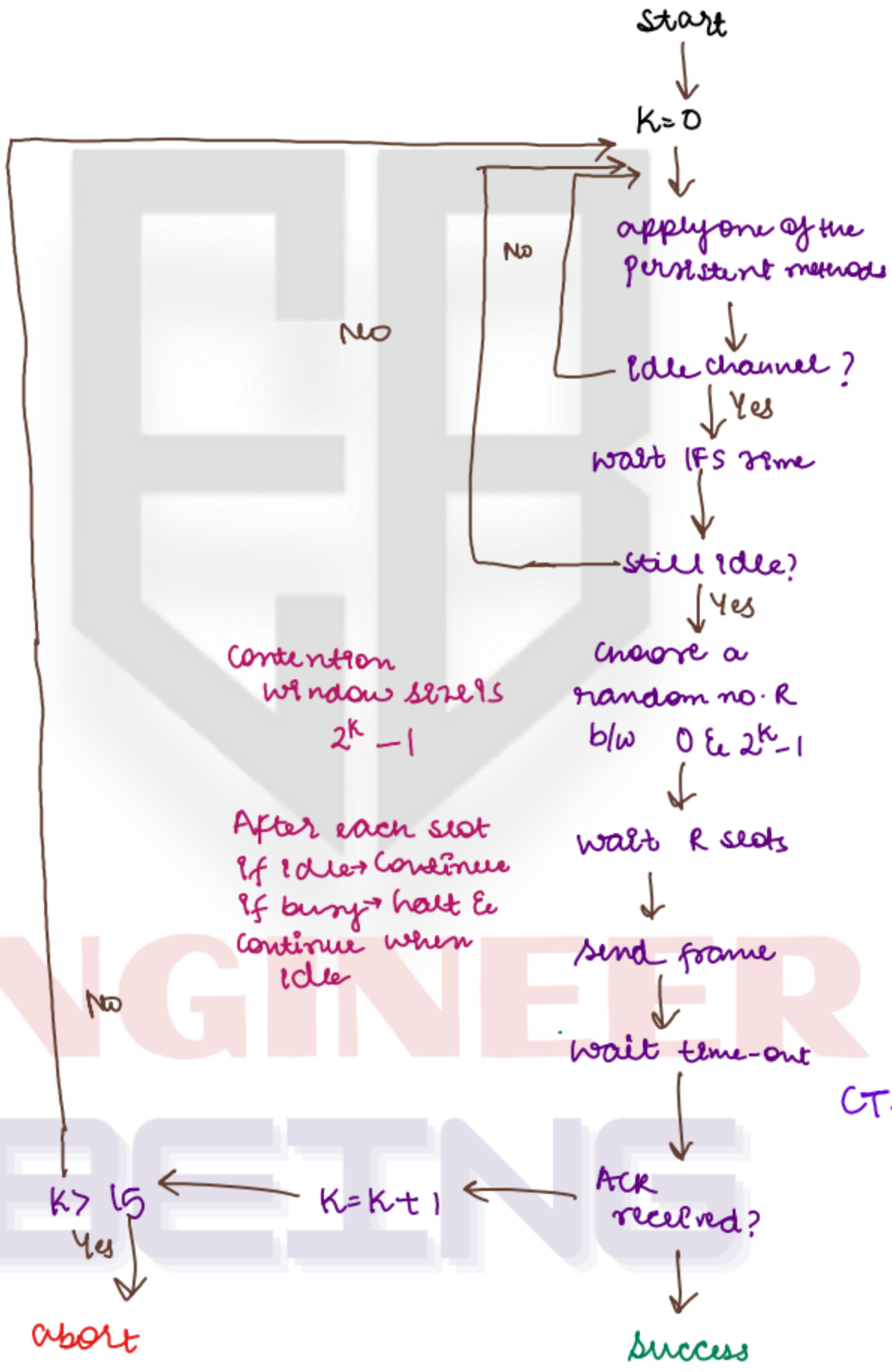
CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is a network access control method used in wireless networks such as Wi-Fi. The purpose of CSMA/CA is to avoid collisions between network devices trying to transmit data at the same time, as well as to manage the shared medium of the wireless network.

The basic idea behind CSMA/CA is similar to CSMA/CD, in that devices listen to the network to check whether the medium is busy before transmitting data. However, in wireless networks, collisions are more difficult to detect due to the nature of wireless communication. As a result, CSMA/CA also includes a collision avoidance mechanism.

Before transmitting data, a device using CSMA/CA sends a request to transmit called a **Request to Send (RTS).** If another device is currently transmitting, it will not respond to the RTS. If the medium is free, the device receiving the RTS will send a **Clear to Send** (CTS) message to indicate that it is ready to receive data. The original device then sends its data.

During the transmission, other devices on the network will not transmit data to avoid collisions. After the transmission is complete, the receiving device sends an **acknowledgement (ACK)** message to confirm that the data was received. If the transmitting device does not receive an ACK, it will assume that a collision has occurred and will retransmit the data after a random amount of time.





ENGINEER BEING

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CONTROLLED ACCESS PROTOCOL

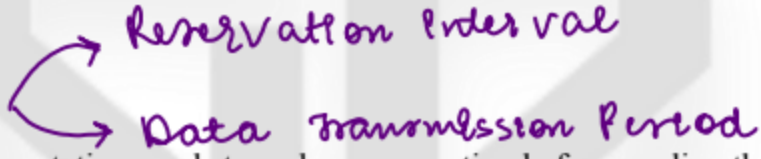
In the Controlled access technique, all stations need to consult with one another in order to find out which station has the right to send the data.

- The controlled access protocols mainly grant permission to send only one node at a time; thus in order to avoid the collisions among the shared mediums.
- No station can send the data unless it has been authorized by the other stations.

The protocols lies under the category of Controlled access are as follows:

- **Reservation**
- **Polling**
- **Token Passing**

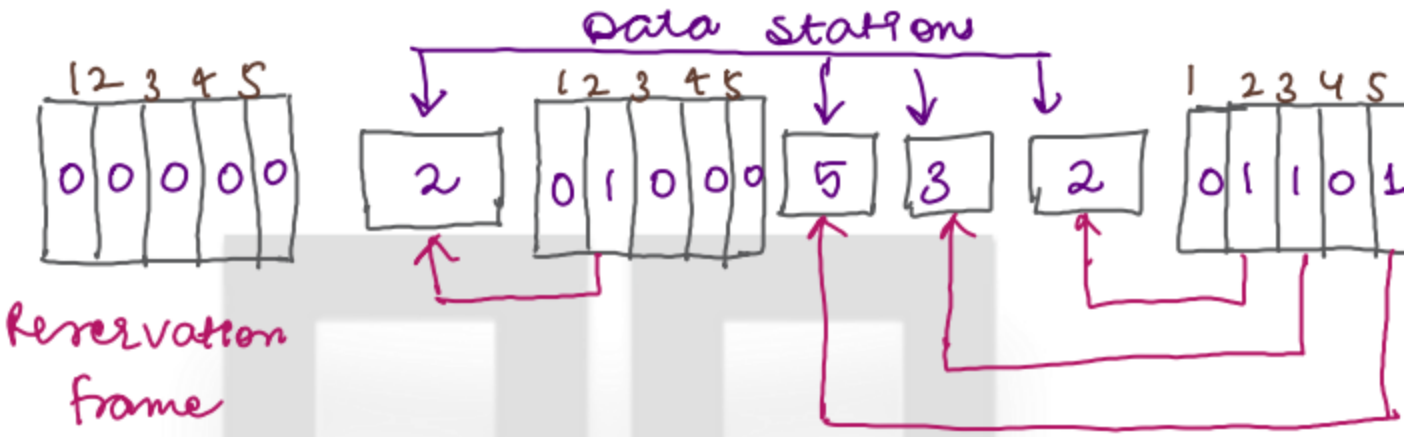
RESERVATION



- In this method, a station needs to make a reservation before sending the data.
- Time is mainly divided into intervals.
- Also, in each interval, a reservation frame precedes the data frame that is sent in that interval.
- Suppose if there are 'N' stations in the system in that case there are exactly 'N' reservation minislots in the reservation frame; where each minislot belongs to a station.
- Whenever a station needs to send the data frame, then the station makes a reservation in its own minislot.
- Then the stations that have made reservations can send their data after the reservation frame.

Example

Let us take an example of 5 stations and a 5-minislot reservation frame. In the first interval, stations 2,3 and 5 have made the reservations. While in the second interval only station 2 has made the reservations.

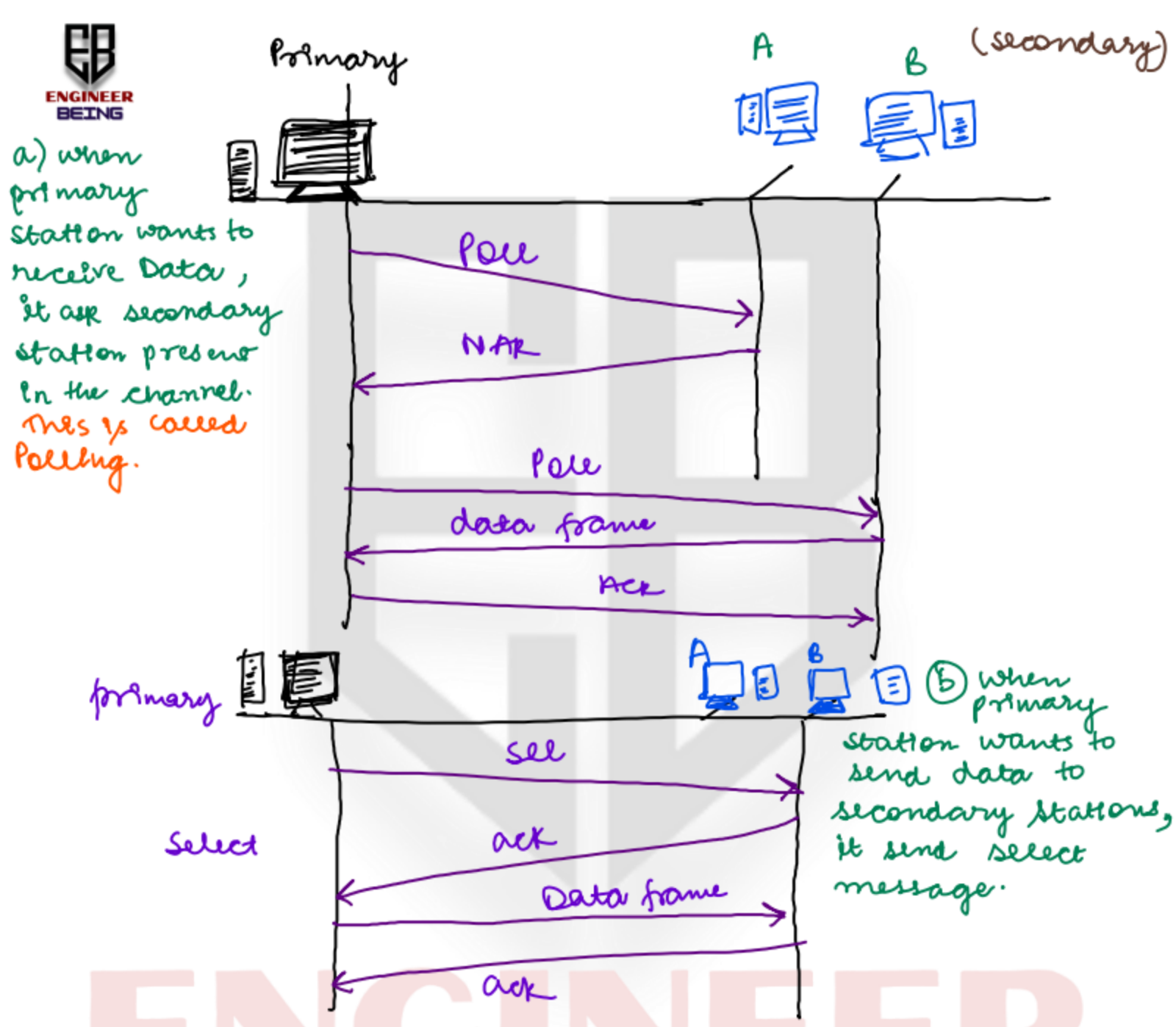


POLLING

- The polling method mainly works with those topologies where one device is designated as the primary station and the other device is designated as the secondary station.
- All the exchange of data must be made through the primary device even though the final destination is the secondary device.
- Thus to impose order on a network that is of independent users, and in order to establish one station in the network that will act as a controller and periodically polls all other stations is simply referred to as polling.
- The Primary device mainly controls the link while the secondary device follows the instructions of the primary device.
- The responsibility is on the primary device in order to determine which device is allowed to use the channel at a given time.
- Therefore the primary device is always an initiator of the session.

WORKING

- When the primary device is ready to receive the data then it must ask(poll) each secondary device in turn if it has anything to send.
- If the secondary device has data to transmit then it sends the data frame, otherwise, it sends a negative acknowledgement (NAK).
- After that in case of the negative response, the primary then polls the next secondary, in the same manner until it finds the one with the data to send. When the primary device received a positive response that means (a data frame), then the primary devices reads the frame and then returns an acknowledgment (ACK)frame



Efficiency

Suppose the time for Polling is Tpoll and Tt will be the time needed for the transferring of data. Then

$$Eff = \frac{Tt}{Tt + TPoll}$$

TOKEN PASSING

There is a special bit pattern or a small message known as a token. It usually passes in predefined order from one station to another station.

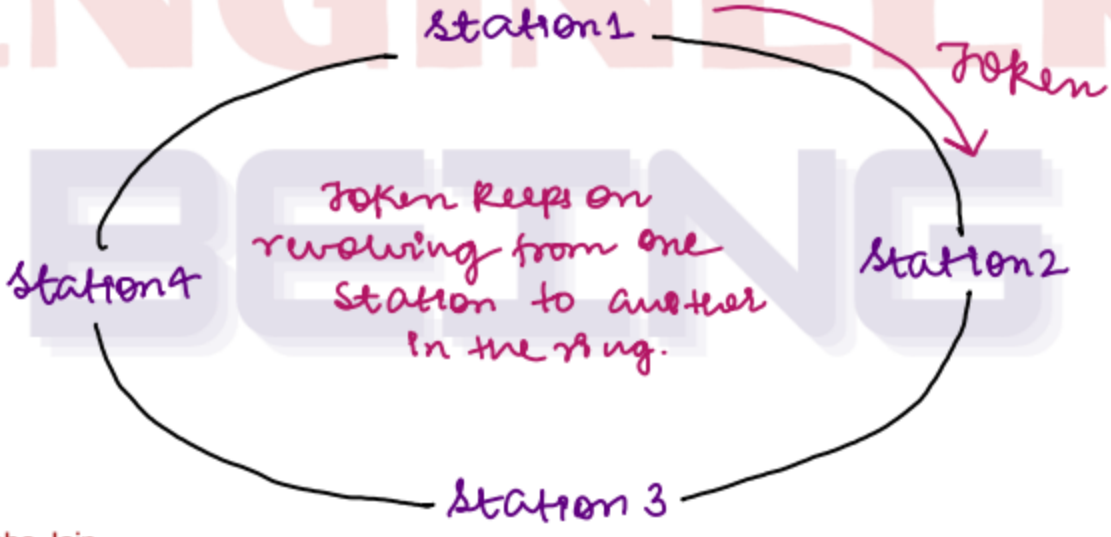
- The station that has this token now has the right to access the channel and can also transmit data present in its queue.
- Any station needs to wait for some time till it gets a token from its predecessor station if it wants to transfer any data.
- After this, the station receives a token and then holds it and has the right to access the channel and transfer its data.
- In the condition, when the station has nothing to transfer then it releases the token soon and passes this token to the next logical station present in that particular ring.
- Although, the station can transfer the data again only when it has a token and it is possible now in the second round only.
- Station simply passes the token to the next logical station if it has nothing to transfer at that particular time.

Advantages

- It resolves the problem of the collision of data on a particular medium.
- It allowed equal access to resources to each connected station in a particular network.

Disadvantages

- Station had to wait for the token even if no other computer wanted to transmit the data.



Performance

There are two parameters through which the performance of the token ring can be identified -

- Delay
- Throughput

Delay: Delay is considered as the measurement of time. Delay is the time difference between when the data packet is ready to transfer and when it is transferred. Therefore, a/N is the average time needed for sending a token to the next station.

Throughput: There is always some traffic on the network in which you transfer the data packet. So the measure of successful traffic on the communication medium is known as throughput.

Throughput $S = \frac{1}{(1 + a/N)}$, for $a < 1$

here N = no. of station, $a = \frac{T_p}{T_t}$,

T_p = propagation delay T_t = Transmission delay

CHANNELIZATION PROTOCOLS

- FDMA (Frequency Division Multiple Access)
- TDMA (Time Division Multiple Access)
- CDMA (Code Division Multiple Access)

These are used in the data link layer to allow multiple users to share a common communication channel efficiently. These protocols are used in wireless networks and other communication systems where multiple users need to access the same physical communication medium.

FDMA divides the frequency band into several non-overlapping sub-channels, each of which is assigned to a specific user. Each user is allocated a different frequency band, and only that user is allowed to transmit data over that band. FDMA is commonly used in analog cellular networks and some satellite communication systems.

TDMA divides the time axis into several time slots, and each user is allocated a specific time slot during which they can transmit data. In TDMA, users share the same frequency band, but they transmit at different times. This protocol is commonly used in digital cellular networks such as GSM (Global System for Mobile Communications) and some satellite communication systems.

CDMA uses a unique code to identify each user's transmission, allowing multiple users to transmit simultaneously over the same frequency band. In CDMA, each user's data is multiplied by a unique code before transmission, and the receiver uses the same code to extract the user's data from the received signal. This protocol is commonly used in digital cellular networks such as CDMA2000 and WCDMA (Wideband Code Division Multiple Access).

All of these protocols allow multiple users to share a common communication channel efficiently, but they differ in the way they allocate the channel to different users.

